

NEURAL BELLMAN OPERATORS: PRESERVATION SUPPLEMENT TO REVISION R10

QIAN QI
Peking University

The current paper and proofs are in ECTA_R10. This supplement preserves the complete completed R9 paper and the entire R9 preservation supplement without deleting equations, numerical tables, counterexamples, or unsuccessful calculations. Earlier precision statements are dated results, not current R10 claims. All underlying historical sources remain unchanged in the revision branch.

READING MAP

The R10 paper contains the affine-preference dual, its complete verification proof, the original-model certificates below .01, the quantitative output-error allowance, the economic cost comparisons, and every main R9 theorem or its explicitly preserved appendix counterpart. A separate point-by-point response to the R9 referee accompanies the paper in the R10 revision directory.

The following pages reproduce the completed R9 paper, including its older .10634 flexible-policy bound. They are followed by the complete R9 supplement, which preserves R8, R4, the delivered R5 numerical material, recursive utility, sophisticated temporal selves, games, stochastic-trace identities, finite safeguards, and historical diagnostics. Historical claims must be read with their original version labels. The R10 current status is determined by its own theorem, result files, and source/result/build manifest, not by the older precision flags reproduced here.

PART I: COMPLETED R9 PAPER, PRESERVED IN FULL

We study neural policy evaluation, improvement, and independent verification in a continuous economy with costly preference adjustment. Feasible learned policies and market-deflator witnesses certify a positive value of access to adjustment under the original utility and first-exit contract. At the central state, the certified gain is at least .04094994 when the adjustment-cost coefficient is two. Constructive interval refinement encloses the no-adjustment optimum within .00905. The flexible policy is evaluated much more precisely than its current .10634 regret bound. For coupled nonconvex controls, a scalar dual branch-and-bound procedure certifies continuous actions through 128 dimensions, and a Cole–Hopf relaxation supplies absolute value lower bounds. A frozen, symmetrically tuned, multi-budget comparison retains all holdout outcomes, including a stronger independently specified feedback baseline. Complete proofs separate policy evaluation, optimality, computational cost, and the economic inequalities each calculation establishes.

KEYWORDS: Stochastic control, Neural policy iteration, Endogenous preferences, Recursive utility, Verification, Computational economics.

1. INTRODUCTION

numerical decision rule is useful for economic analysis only when its error is small relative to the comparison being made. This requirement is distinct from a small training loss. It is also distinct from outperforming another learning algorithm. When a decision maker can change a reference state, an apparent welfare gain may reflect either the adjustment technology or an accurate evaluation of the policy without that technology. When termination is an economic contract, an inaccurate treatment of first exit can dominate both effects.

This paper studies Neural Bellman Operators (NBO): separated policy evaluation and feasible policy improvement, implemented by neural networks and assessed by independent verification. A motivating economy combines consumption, portfolio choice, stochastic risk aversion, and costly preference adjustment. The analysis retains its original running utility, continuous controls, correlated Brownian shocks, and first-exit liquidation covenant. Neither reflecting the price at a training boundary nor altering the running payoff is used to establish the economic results below.

The main economic result is a strictly positive, verified lower bound on the value of access to preference adjustment. At the central initial state and adjustment-cost coefficient $k = 2$, access to optimal welfare is by at least .04094994 in the model’s fixed utility units. The corresponding upper bounds are .06274788 at $k = .5$ and .00781212 at $k = 8$. These are comparisons with the *central* no-adjustment economy, not merely with an arbitrarily chosen no-adjustment policy. A feasible learned policy supplies the flexible economy’s lower bound. An auxiliary market-deflator witness supplies the restricted economy’s upper bound while allowing every original utility-consumption and portfolio policy. A scalar state used to construct the witness does not change the economic opportunity set.

The computational argument has three parts. First, smooth one-sided witnesses bound the final stopped problem without assuming a smooth optimal value or regularity of the boundary for every control. Second, fitting and certification are separate operations. A polynomial fitted continuous-generator equation is converted to outward Bernstein coefficients and refined

to evaluate a trained time-control policy under the exact continuous stopping contract. bounded-source Feynman–Kac construction and exponential localization potentials turn adjustment market-deflator relaxation into a verifiable economic upper bound. Weighted boxes refine its initial value enclosure from width 1.0068 to below 0.000120. The distinction between evaluation accuracy and optimality accuracy matters. At $k = 2$ the payoff learned policy has a value enclosure of width 1.33×10^{-5} , but its current regret bound is .10634, not .0000133. This replaces the previous 8.33956 bound for a different policy with a substantially smaller original-model certificate; it does not establish the .01 full-policy target. The positive access-welfare result is valid nevertheless because the feasible flexible policy with a substantially tighter upper bound on the restricted π . The paper therefore establishes a specific economic inequality rather than interpreting solved approximation as an optimal policy surface.

Second contribution addresses the inner optimization needed for verification. In a concave control family, all nonconvex coupling depends on the sum of the actions. A dimensional branch-and-bound calculation, together with strongly convex scalar dual problems, certifies the full continuous action box without a tensor action grid. Executed tests reach gaps below 10^{-5} through 128 action dimensions. At the eight- and sixteen-dimensional learned-critic test states, interval differentiation of the actual network weights also bounds the error in the floating-point critic gradient. This is an implemented bridge between parameters and an action certificate, not an assumption that an optimizer’s stationary policy is globally greedy.

For the same nonconvex family is given an absolute accuracy axis. A pointwise running-cost minimization and removal of control bounds admit a Cole–Hopf representation that retains the nonquadratic terminal payoff. Orthogonal Gaussian coordinates integrate analytically, reducing a positive scalar integral with an outward enclosure. The resulting number is a lower bound for the unchanged original minimization problem. Relative performance is evaluated empirically in a frozen holdout experiment: three dimensions, two training budgets, twelve seed parameters per panel, equal learning-rate search budgets, warmed initialization, randomized method of moments, and a third, independently specified feedback rule. The pinned author implementation of CartNet is not modified.

For local verification, monotone approximation, and primal–dual numerical methods supply theoretical foundations. The stopped Itô argument is not claimed as a new general verification theorem. Nor is alternating neural evaluation and improvement claimed as a new algorithm. The contribution is the constructive combination of original-contract verification, independent economic upper and lower bounds, computable neural-jet allowances, and a non-anticipative global action audit, with each error quantity matched to an executed calculation. Section 8 compares these objects theorem by theorem with neighboring methods. Recursive utility, sophisticated temporal selves, dynamic games, stochastic-trace identities, counterexamples, and complete preceding manuscripts are preserved in the supplement rather than being used to change the empirical claims of the present application.

2. THE ECONOMIC ENVIRONMENT

The state is $s = (u, x)$, where u denotes relative risk aversion and x denotes wealth. An admissible action is $a = (c, \theta, p)$ in the compact set

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

backups and time controls actually evaluated below are bounded and satisfy this requirement. More termination,

$$du_t = \theta_t dt + .05 dZ_t^u, \quad (1)$$

$$dx_t = [(.02 + .06p_t)x_t - c_t]dt + .2p_tx_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

horizon is $T = 1$, discounting is $\rho = .04$, and the open state domain is $D = (1.2, 2.8) \times (0, 2)$. Write $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. For $k > 0$, running utility and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

policy payoff and optimal value are

$$J_k^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k = \sup_\pi J_k^\pi. \quad (5)$$

no-adjustment value V_R restricts θ to zero and retains *all* admissible consumption and portfolio choices. It is independent of k . The cost experiment holds the controls, transition law, stopping rule, settlement, utility normalization, and consumption units fixed. A state-dependent transformation of utility would generally define another economy.

The early settlement fee is a liquidation covenant. There is no continuation consumption or wealth injection following exit. At the lower wealth boundary the largest admissible wealth is negative. At the upper boundary, $p = 0$ eliminates the wealth diffusion and permits an upward drift. Thus neither a silently viable rectangle nor a theorem requiring uniform controlled-boundary regularity can be substituted for this contract. The verification below requires only local regularities for smooth test functions on the trace at which the process is actually stopped.

For a smooth function v , put

$$\mathcal{L}^a v = \theta v_u + [(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$R_a v = v_t + \mathcal{L}^a v + \ell_k - \rho v. \quad (7)$$

The mixed derivative is retained. Its sign is not inferred solely from the negative shock correlation.

PROPOSITION 1—Complete selection at a fixed jet: *For the generator in (7), a global maximizing action can be selected componentwise. Consumption is $\text{proj}_{[.05,.8]}(v_x^{-1/u})$ when $h_1 > 0$ and is $.8$ otherwise. Adjustment is $\text{proj}_{[-.2,.2]}(v_u/k)$. Let $h_1 = .06xv_x - .0025xv_{ux}$ and $h_2 = .04x^2v_{xx}$. If $h_2 < 0$, the portfolio is $\text{proj}_{[-.5,.8]}(-h_1/h_2)$; if $h_2 \geq 0$, an endpoint maximizing $h_1p + h_2p^2/2$ is selected, with a fixed tie rule.*

The statement includes convex and zero-curvature portfolio objectives and nonpositive marginal wealth value. It applies to the continuous generator, not to the interpolated, killed-Euler approximation. It says nothing about global optimization over shared neural parameters.

fixed policy, exact evaluation solves $R_\pi v = 0$ with the stopping contract. Exact improvement maximizes the Hamiltonian of that fixed evaluation. Denote these operators by $\mathcal{E}$ and $\mathcal{I}$; policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. A neural implementation approximates both operators, independent evaluator measures the payoff of the policy that will actually be used.

In raw NBO implementation, critic updates hold the actor fixed. Actor updates hold parameters fixed, but retain differentiation through the controlled successor state. A parameters, when present in a game, are also frozen during unilateral improvement. This prevents an actor loss from lowering itself by changing the object being evaluated. It makes the joint network optimization convex. The ideal policy-iteration compactness and bounding arguments are retained in the supplement; no such argument is invoked to certify the learned Adam trajectory.

Inherited finite safeguard is a distinct algorithm. For $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, positive sub-stochastic P_n^a , and $\delta = e^{-\rho\Delta}$, define the policy-evaluation residual $e_n = \|v_n - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$ and candidate improvement gap $\eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$. If the terminal error is at most b , backward comparison yields

$$\|V_{h,n}^* - V_{h,n}^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

The ρ -dependent envelope and threshold-correction proof are reproduced in the supplement. The audit can reduce the fraction of decisions changed while leaving the cost of a complete audit essentially unchanged. In particular, the retained 125-action tensor audit is not an enumerative certifier of Section 6. Its floating-point envelope is not relabeled as an continuous-time bound.

3.1. A quantitative bridge from a fitted network

$\rho(h) = (1 - e^{-\rho h})/\rho$, with $C_0(h) = h$.

ASSUMPTION 1—Admissible witnesses: *Witnesses are bounded, continuous on the stopped trace and $C^{1,2}$ in the interior of each time slab. Localized Itô identities pass to the stopping time. Uniform integrability. At slab interfaces an upper witness has nonpositive forward jumps and a lower witness has nonnegative forward jumps. The implemented policy has a well-defined state equation.*

LEMMA 1—Stopped one-sided comparison: *Suppose Assumption 1 holds, $U \geq g - b_U$ and b_L on the stopped trace, $\sup_a R_a U \leq \epsilon_U$, and $R_\pi L \geq -\epsilon_L$, where all four allowances are negative. Then*

$$J_k^\pi \geq L - b_L - C_\rho(T-t)\epsilon_L, \quad V_k \leq U + b_U + C_\rho(T-t)\epsilon_U, \quad (9)$$

$$0 \leq V_k - J_k^\pi \leq U - L + b_U + b_L + C_\rho(T-t)(\epsilon_U + \epsilon_L). \quad (10)$$

These inequalities do not require an attained optimizer or a smooth optimal value.

LEMMA 2—Cover-and-oracle certificate for neural output: *Let a continuous neural witness and a feasible neural policy π satisfy Assumption 1, with stopped-trace error at most b . Let*

ulation proves, throughout cell j ,

$$|R_\pi v(z) - R_\pi v(z_j)| \leq \omega_j, \quad |\Gamma(z) - \Gamma(z_j)| \leq \nu_j, \quad \Gamma = \sup_a R_a v - R_\pi v.$$

6

pose the representative evaluation intervals imply $|R_\pi v(z_j)| \leq e_j$ and a global action oracle satisfies $\Gamma(z_j) \leq q_j$, including jet, inner-solve, and arithmetic allowances. Set $e = \max_j (e_j + \omega_j)$ and $q = \max_j (q_j + \nu_j)$. Then

$$0 \leq V_k - J_k^\pi \leq 2b + C_\rho(T - t)(2e + q). \quad (11)$$

particular, certified derivative bounds M_j, N_j and cell radii h_j permit $\omega_j = M_j h_j$ and $\nu_j = N_j h_j$. Any sequence of learned outputs for which these verified allowances vanish has vanishing regret, irrespective of whether its parameters converge.

This is a quantitative a posteriori connection to network errors, not a claim that an empirical mean-square loss controls a supremum. Interval differentiation or weight-based derivative bounds can supply the cell moduli. They may be conservative, and covering a high-dimensional domain remains costly. Section 6 executes the jet-to-action part for actual learned critics. An economic calculation instead fits an independent polynomial witness; it does not falsely claim to have certified a whole-state neural critic merely because its action audit passed at selected states.

4. CONSTRUCTIVE CERTIFICATES IN THE ORIGINAL ECONOMY

All initial numerical bounds in this section concern $(t, u, x) = (0, 2, 1.25)$. A central-state certificate is not advertised as a uniform certificate for every state in D . The policy used here was newly trained in a time-control neural subclass. The retained lattice-trained, state-feedback policies, their finite corrections, and their previous continuous bounds remain separately available in the supplement.

4.1. Fitting an upper witness without importing an optimal value

At $h = 1 - t$ and restrict the *verification domain*, not the economic domain, to $u \in [1.5, 2.6]$.

$$F(h, u) = h \sum_{i=0}^{12} \sum_{j=0}^{14} \beta_{ij} T_i(2h - 1) T_j \left(2 \frac{u - 1.5}{1.1} - 1 \right).$$

The coefficients are obtained from a method-of-lines initializer and residual least squares. The initializer's collocation residual is not the certificate. For $w = G + F$, the consumption and portfolio maximizers are .8 and .8. Maximizing the remaining wealth term over $x \in [.5, 2]$ gives a scalar residual majorant

$$\mathcal{R}_k F = -F_h + .00125 F_{uu} - .04 F + H_k \{F_u - .04(u - 2)\} + \frac{.8^{1-u}}{1-u} \quad (12)$$

$$+ .0008(u - 2)^2 + .00195 + .00352 - .04 - .004 \log 2, \quad (13)$$

$$H_k(q) = \max_{|\theta| \leq .2} (q\theta - k\theta^2/2).$$

coefficients are enclosed at 70-digit interval precision and converted to Bernstein polynomials. The utility term is replaced by a polynomial with a proved uniform remainder below 10^{-9} . De Casteljau subdivision uses outward rounding at every elementary step. The positive-saturation, negative-saturation, and interior-branchers of KATONs are included whenever a specified q range permits them. Outward thresholds also cover a cell touching a switching boundary. The same calculation proves $F/h \geq -8$.

POSITION 2—Localized fitted upper witness: *Suppose $F(0, u) = 0$, $F/h \geq -8$, and ϵ on the verification rectangle. For $k \geq .5$, let*

$$Z(h, u) = 8h\{e^{-120(u-1.5)+42h} + e^{-160(2.6-u)+64h}\}, \quad (14)$$

$$U(t, u, x) = G(u, x) + F(1-t, u) + Z(1-t, u) + \epsilon_+ C_\rho(1-t). \quad (15)$$

$U(0, 2, 1.25) \leq U(0, 2, 1.25)$. No reflecting boundary or truncation of the original policy is used at $u = 1.5$ or 2.6 .

Exponential terms bound continuation when an artificial verification face is reached. The valid upper witness G is used beyond that face. At the central state the localization is less than 1.219×10^{-7} , rather than an eight-unit initial gap. Uniform residual on and localization are different terms and are reported separately.

4.2. A learned feasible policy and independent stopped evaluation

Factor has one time input, a twelve-unit tanh layer, and two outputs. At sixteen fixed nodes, a sigmoid constrains adjustment to $[0, .2]$, and a differentiable budget projection constrains consumption to $[\cdot05, .8]$. The portfolio is zero. Training uses the original expected utility, Gaussian quadrature, and no optimal-value or greedy-action labels. The deployed policy is the resulting frozen binary64 constants on each time slab; there is no claim that this control class exhausts the state-feedback opportunities in (5).

Budget projection enforces

$$\sum_{j=0}^{15} c_j \int_{j/16}^{(j+1)/16} e^{-\cdot02r} dr < 1.25 - \cdot5e^{-\cdot02}. \quad (16)$$

$= e^{-\cdot02t} \{1.25 - \int_0^t e^{-\cdot02r} c_r dr\}$ is deterministic. Since consumption exceeds $\cdot02(1.25)$, wealth decreases and its interval-enclosed terminal value remains strictly above $\cdot5$. Wealth never exits the remaining preference exit is controlled by a Brownian maximal inequality, not by a reflecting boundary check.

Evaluation independent of the training quadrature approximates utility by a degree-90 polynomial in $u - 2$, with a uniform analytic remainder on the original preference interval. Free moments are evaluated by recurrence, integrated over outward time rectangles, and used for the stopped event using a maximal probability bound and Cauchy–Schwarz. The utility, wealth, adjustment expenditure, polynomial approximation, time integration, exact-exit test, and arithmetic are all enclosed. There is no Euler, action-grid, or spatial-interpolation on this route. Those omissions are consequences of the deployed policy and analytic proof, not uncomputed entries assigned zero.

I reports the actual original-economy bounds. Table II makes the constructive refinement explicit. The last row gives substantially more precise policy evaluation than optimality

k	Policy payoff enclosure	Optimal-value upper	Regret upper
0.5	$[-1.27324522, -1.27322724]$	-1.16618072	0.10706450
2	$[-1.29504316, -1.29502986]$	-1.18870455	0.10633861
8	$[-1.32818098, -1.32817412]$	-1.22254633	0.10563464

off endpoints and upper bounds are rounded outward. Each regret upper is the independent optimal-value upper minus the lower endpoint of the learned policy's payoff interval. All controls in the optimal-value bound are continuous. The .01 full-policy target is not met in any row.

TABLE II
EXECUTED REFINEMENT FOR THE $k = 2$ CERTIFICATE

Time boxes	Payoff width	Sec.	Bernstein boxes	Residual upper	Sec.	Regret upper
1024	0.000209	0.363	36	0.00477	0.079	0.110237
4096	5.25e-05	0.614	57	0.00195	0.116	0.107396
16384	1.33e-05	1.618	86	0.000894	0.186	0.106339

boxes refine the independent policy evaluator. Bernstein boxes refine the already fitted upper witness. The two nds columns measure the respective integration and subdivision stages; complete timings, including coefficient construction, are in the raw records. A shrinking evaluation or residual allowance need not eliminate the approximation gap between the chosen upper witness and policy.

fication. The fitted witness is independent of the policy and valid over the full original th and action ranges. Its loss of wealth-specific information is economically consequential. parate wealth-sensitive linear-programming pilot is retained with its computation log and cted candidate diagnostics; an unvalidated collocation objective is not used in either table.

5. CERTIFYING THE VALUE OF PREFERENCE ADJUSTMENT

for each feasible policy write $J_k(\pi) = A(\pi) - kB(\pi)$, where

$$B(\pi) = \mathbb{E}^\pi \int_t^\tau e^{-\rho(r-t)} \theta_r^2 / 2 \, dr, \quad 0 \leq B(\pi) \leq .02C_\rho(T-t).$$

policy set is common to every k .

THEOREM 3—Accuracy-calibrated cost comparisons: *The optimal value is convex and increasing in k . If optimizers exist, their adjustment budgets are nonincreasing across nct prices. More generally, let π_i be ε_i -optimal at k_i , with $k_2 > k_1$. Then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (17)$$

($k \in [\underline{V}(k), \bar{V}(k)]$), the budget of an optimizer at k , for $k_- < k < k_+$, lies in the primitive get range intersected with

$$\left[\frac{\underline{V}(k) - \bar{V}(k_+)}{k_+ - k}, \frac{\bar{V}(k_-) - \underline{V}(k)}{k - k_-} \right]. \quad (18)$$

he theorem concerns discounted adjustment expenditures, not a pointwise ordering of rols. A re-solved V_u/k is not a comparative-static proof. The central full-value enclosures

5.1. A market-deflator witness for the restricted optimum

NBO: R10 PRESERVATION SUPPLEMENT

9

inter comparison becomes possible by exploiting the restricted economy's structure. The auxiliary process

$$dY_t = .02Y_t dt - .3Y_t dZ_t^x, \quad Y_0 = 1.55. \quad (19)$$

st-process constructed from the original shocks. It does not restrict portfolio information, wealth, or add a traded asset. For

$$W_0(u, x, y) = y(x - .5) + .1 \log(.5) - .02(u - 2)^2,$$

ms in the generator of Yx cancel exactly. With $\theta = 0$, define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad (20)$$

$$S(u, y) = \Psi(u, y) + .01y - .00005 - .004 \log(.5) + .0008(u - 2)^2. \quad (21)$$

1.5, 2.6], $y \in [.2, 12]$, one has $-8 < S < 0$. Extend S to a bounded function $\tilde{S} \in [-8, 0]$ ing its two arguments at these verification faces. Let

$$F_R(h, u, y) = \mathbb{E}_{u,y} \int_0^h e^{-.04r} \tilde{S}(U_r, Y_r) dr, \quad (22)$$

$U_r = .05 dZ_r^u$ and Y follows (19). Thus $-8h \leq F_R \leq 0$.

LEM 4—A restricted-economy upper bound: Let Z_R be the sum of the following tive localization potentials:

$$Z_{R,u} = 8h \{ e^{-200(u-1.5)+50h} + e^{-240(2.6-u)+72h} \}, \quad (23)$$

$$Z_{R,y} = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (24)$$

$$V_R(0, 2, 1.25) \leq W_0(2, 1.25, 1.55) + F_R(1, 2, 1.55) + Z_R(1, 2, 1.55). \quad (25)$$

er bound covers all adaptive original consumption and portfolio controls with $\theta = 0$. uently, for any feasible flexible policy with certified payoff lower bound $\underline{J}_k$ and any ound $\bar{V}_R$ from (25),

$$V_k(0, 2, 1.25) - V_R(0, 2, 1.25) \geq \max\{0, \underline{J}_k - \bar{V}_R\}. \quad (26)$$

xpectation in (22) has an explicit two-Gaussian representation. We transform time by evaluate constant and linear Taylor terms with interval-enclosed weighted moments, and the Hessian remainder on each box. Exponential Taylor remainders, Gaussian tails, ilization are included. This produces the executed refinement in Table III; it does not o a mesh difference as an error estimate. The final restricted-optimum upper bound

	Weighted boxes	Dual witness-value enclosure	Width	Seconds
10	16384	$[-1.33876765, -1.33309541]$	0.00567	2.67
	131072	$[-1.33676857, -1.33544680]$	0.00132	15.13
	1048576	$[-1.33631272, -1.33599309]$	0.00032	110.24

se intervals enclose the *dual witness value*, not the restricted optimum. Only their upper endpoints are used as upper bounds for that optimum. A separately evaluated restricted policy gives the primal lower bound. Original stopping and all adaptive consumption and portfolio opportunities are retained.

TABLE IV

CERTIFIED ORIGINAL-MODEL EXPENDITURES AND VALUE OF ACCESS

k	Policy adjustment budget	Access gain: lower	Access gain: enclosure
0.5	$[0.01960483159556, 0.01960483159557]$	0.06274788	$[0.06274788, 0.17885870]$
2	$[0.01242015505497, 0.01242015505498]$	0.04094994	$[0.04094994, 0.15633486]$
8	$[0.00192618262027, 0.00192618262028]$	0.00781212	$[0.00781212, 0.12249308]$

expenditure column evaluates the displayed learned policy. The access columns bound the difference between the *o* optimal values. The lower endpoint pairs a feasible flexible policy with the restricted dual upper; the upper endpoint pairs the flexible upper with a feasible restricted policy. The positive lower endpoints are informative economic guarantees even though the two-sided access intervals are not narrow estimates of the welfare magnitude.

1.33599309 after outward display rounding. The evaluated restricted policy has payoff $[-1.34503942, -1.34503585]$, so the restricted problem itself has an initial optimal-value enclosure narrower than .00905.

the welfare lower bounds remain strictly positive at all three prices. At $k = 2$, the quadrature enclosure of the restricted witness is more than two orders of magnitude narrower than the certified access gain. This comparison establishes a minimum economically meaningful gain without assuming that the flexible learned policy is optimal. It does not identify the exact access, an optimal adjustment-budget magnitude, or an optimal pointwise policy ordering.

6. GLOBAL ACTION CERTIFICATION WITHOUT A TENSOR GRID

Consider the following minimization problem, which is also used for the external comparison:

$$dX_t = 2a_t dt + \sqrt{.32} dW_t, \quad X_0 = 0, \quad a_t \in [-1, 1]^d, \quad (27)$$

$$f_d(a) = \frac{1}{d} \sum_{i=1}^d \{a_i^2 + .1 \sin^2 a_i\} + .3 \sin \left(\sum_i a_i \right), \quad (28)$$

$$q_d(x) = \frac{1}{d} \sum_{i=1}^d (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right), \quad (29)$$

$$V_d^* = \inf_a \mathbb{E} \left[\int_0^1 f_d(a_t) dt + q_d(X_1) \right].$$

objective has genuinely coupled nonconvex actions. No analytic optimal-action or optimal-labels are supplied to training. At a fixed critic gradient p , the action-dependent Hamilto-

tion dim.	Tolerance	Verified gap	Scalar boxes	Convex subproblems	Seconds
8	1e-03	0.000258	33	264	0.299
8	1e-05	6.72e-06	47	376	0.493
16	1e-03	0.000931	37	592	0.476
16	1e-05	4.42e-06	57	912	0.751
32	1e-03	0.00068	85	2720	1.498
32	1e-05	3.52e-06	103	3296	1.991
64	1e-03	0.000647	163	10432	6.041
64	1e-05	3.06e-06	185	11840	5.955
128	1e-03	0.000948	317	40576	19.229
128	1e-05	4.68e-06	339	43392	20.651

an jet is used per dimension, at both tolerances. At $d = 8, 16$, these are gradients of retained learned critics at ded states; displayed gaps include interval neural-derivative allowances. At larger dimensions they are ntly generated jets for the same Hamiltonian family. The experiment measures certificate work for these instances, not an average-case scaling law or a whole-state neural PDE residual.

11

$$H_p(a) = \sum_i f_i(a_i) + .3 \sin z, \quad f_i(s) = \frac{s^2 + .1 \sin^2 s}{d} + 2p_i s, \quad z = \sum_i a_i.$$

is strongly convex with modulus at least $m = 1.8/d$.

LEM 5—Scalar global partition and convex dual bounds: *For any multiplier λ and d], put*

$$D(\lambda) = \sum_i \min_{|s| \leq 1} \{f_i(s) - \lambda s\}.$$

ular interval $z \in [z_0 - r, z_0 + r]$, a lower bound for the full Hamiltonian is

$$D(\lambda) + \lambda z_0 + .3 \sin z_0 - |\lambda + .3 \cos z_0| r - .15 r^2. \quad (30)$$

ible action supplies an upper bound. Each inner convex minimum can itself be bounded $v q(s_0) - e(s_0)^2 / (2m)$, where $q = f_i - \lambda s$ and e is the derivative magnitude in a feasible direction, or the full derivative magnitude at an interior point. Outward versions of unds, followed by scalar subdivision and pruning, give a global gap on $[-1, 1]^d$. eored gradient $\hat{p}$ differs from the true neural gradient by componentwise bounds δ_i , the action's true-gradient gap is at most its stored-gradient gap plus $4 \sum_i \delta_i$.

ng-point root finding only proposes multipliers and actions. It is not trusted to solve vex subproblems exactly. Even an inaccurate proposal remains valid after the strong-ty allowance. This structure removes tensor enumeration; it does not establish a on-free complexity theorem or solve a generic nonseparable nonconvex Hamiltonian. 28-dimensional 10^{-5} calculation uses 339 scalar boxes and 43,392 scalar convex ems. It replaces neither raw actor training nor the preceding finite tensor safeguard in ct. At the learned eight- and sixteen-dimensional jets, interval forward differentiation eored critic weights bounds the ℓ^1 gradient discrepancies below 4.584×10^{-17} and

d	Quadrature cells	Relaxed-value enclosure	Original-value lower	Seconds
8	1024	$[-0.08599082, -0.08384626]$	-0.08599082	0.260
8	4096	$[-0.08518701, -0.08465086]$	-0.08518701	1.081
8	16384	$[-0.08498598, -0.08485194]$	-0.08498598	4.506
16	1024	$[-0.20520651, -0.20360335]$	-0.20520651	0.271
16	4096	$[-0.20460665, -0.20420585]$	-0.20460665	1.067
16	16384	$[-0.20445644, -0.20435623]$	-0.20445644	4.212
32	1024	$[-0.28757809, -0.28684340]$	-0.28757809	0.245
32	4096	$[-0.28730394, -0.28712026]$	-0.28730394	0.988
32	16384	$[-0.28723515, -0.28718923]$	-0.28723515	4.172

encloses the relaxed value. Its lower endpoint is also a lower bound for the original bounded-control value. The width shown by successive enclosures is quadrature uncertainty; it does not include or estimate the structural relaxation gap.

7×10^{-17} , respectively. These allowances are propagated into the action gap. Theorem 2 identifies the additional state-cover and residual quantities that would be needed for a whole-policy guarantee.

7. AN ABSOLUTE BOUND AND A FROZEN EXTERNAL COMPARISON

7.1. A rigorous lower bound for the unchanged nonconvex family

Let $r_0 = 1 + .1 \sin^2(1)$, $\sigma^2 = .32$, and $\lambda_d = \sigma^2 r_0 / (2d)$. Since $\sin^2 a \geq \sin^2(1)a^2$ for $|a| \leq 1$, replacing the running cost by $r_0 \|a\|^2 / d - .3$ and allowing unbounded controls gives a relaxation. The original terminal function q_d is retained.

PROPOSITION 3—Scalar-integral absolute lower bound: *Let Z be normal with mean 0 and variance $\sigma^2 / (r_0 + 4)$ and variance $\sigma^2 r_0 / (r_0 + 4)$. Then*

$$V_d^* \geq \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda_d \log \mathbb{E} \exp \left[-\frac{.2}{\lambda_d} \cos Z \right]. \quad (31)$$

The outward enclosure of the positive expectation yields a rigorous numerical lower bound for the original objective (29).

The relaxation is an upper/lower-bound device, not a new training problem with a known manufactured value. The terminal cosine, which couples all state coordinates, remains in the original expectation. Its tails are bounded analytically and its interior is integrated by interval angles.

7.2. Holdout design and exact deployment interpretation

The comparison retains the original actor and critic architectures: two tanh hidden layers of 48 units each, a bounded actor output, and the same terminal-value embedding. NBO uses separated evaluation and improvement with Adam, three critic steps per actor step, 256 training states, and four successors. SOC-MartNet uses its pinned author solver, RMSprop, a 500-output test network, and the inherited $J = 2$, $K = 1$ multiplier setup. These different training systems are disclosed rather than described as identical implementations. The committed protocol freezes dimensions 8, 16, 32 and training budgets of 10 and 30 seconds. At each dimension, both methods receive the same six validation trials: two seeds

d	Budget	NBO cost	SOC cost	LQ cost	Median difference	Adjusted p
8	10	0.5812	1.0842	0.5121	-0.4783	0.001465
8	30	0.5785	1.0557	0.5121	-0.4783	0.001465
16	10	0.4663	0.7893	0.1740	-0.1672	1
16	30	0.4630	0.8456	0.1740	-0.2878	0.438
32	10	0.3813	1.2011	0.0854	-0.8691	0.001465
32	30	0.3799	1.0218	0.0854	-0.6962	0.001465

realized cost is better. The median difference is NBO minus SOC-MartNet. The last column is a one-sided sign-test p -value adjusted across the six predeclared dimension–budget panels. The sign test concerns the joint training-and-audit pipeline under independent seed draws, not an outward numerical error guarantee.

TABLE VIII

ABSOLUTE REFERENCE EXCESS AND ACTUAL TRAINING CLOCKS

d	Budget	Lower bound	NBO excess	SOC excess	NBO sec.	SOC sec.
8	10	-0.08499	0.6662	1.1692	10.002	10.006
8	30	-0.08499	0.6635	1.1407	30.004	30.008
16	10	-0.20446	0.6707	0.9938	10.002	10.005
16	30	-0.20446	0.6674	1.0500	30.004	30.006
32	10	-0.28724	0.6686	1.4883	10.003	10.007
32	30	-0.28724	0.6672	1.3090	30.003	30.007

is the estimated mean policy cost minus the rigorous original-value lower bound. It provides an absolute axis but is not itself a deterministic regret enclosure: sampling error and floating-point rollout evaluation are not certified. Training clocks include completed-update overshoot and exclude separately recorded warmup and setup.

learning-rate multipliers. Selection minimizes mean independent validation cost, subject to a fixed tie rule. Twelve different holdout seeds, 720–731, are then evaluated at each dimension–budget pair. Within each pair, actor and critic starting weights are identical and their gradients are checked. Framework warmup precedes both clocks, and method order is seed-randomized. Training-only and setup-plus-training clocks are retained. The baseline $a_i(t, x) = \frac{1}{1 + 4(1 - t)}[-2(x_i - .5)]$ is specified independently from the associated quadratic objective and is evaluated on the unchanged nonquadratic objective.

The trained network is deployed as a sample-and-hold control on 128 dates and evaluated on 96 independent, paired Gaussian paths. In (29), drift under a held action and volatility are constant on a slab; the Gaussian transition and running-cost integral are therefore exact in the limit for that deployed continuous-time policy. A 64-date evaluation uses aggregated increments and changes the deployment policy. Its difference from the 128-date result is bounded by an Euler-error bound. Floating-point network evaluation and Monte Carlo uncertainty are distinct from the outward deterministic lower bound in Table VI.

Holdout results favor NBO over the author implementation in mean cost in all six panels, though the interpretation is not uniform. In dimensions 8 and 32, NBO has lower paired cost than the author implementation at both budgets; each multiplicity-adjusted sign-test value is .001465. In dimension 16 the adjusted values are 1 and .438, so these panels do not establish a robust ordering under the declared test. Tripling the budget gives only small improvements in mean cost, from .5812 to .5785, .4663 to .4630, and .3813 to .3799 respectively. Most notably, the independently specified quadratic-feedback baseline has lower mean cost than the learned methods in every panel. Its mean costs are .5121, .1740, and .0854 in dimensions 8, 16, and 32. These retained results support the stated NBO–SOC-MartNet comparison at the

already reached an accurate optimum.

The raw records retain every seed difference, the median and interquartile range, approximate percent intervals, costs for all three rules, path standard errors, training curves, optimizer clocks, and checkpoints. Approximate Student intervals are descriptive; the exact sign calculation avoids assuming Gaussian seed-level differences. The source protocol predates holdout execution, while the benchmark family and adapters themselves were developed in earlier revisions. Holdout results do not turn one problem family into evidence of general superiority across economic environments. Two budget points are not a matched-accuracy frontier, and a finite absolute lower bound need not be tight enough to rank methods at a small prescribed regret.

8. RELATION TO RIGOROUS CONTROL AND NEURAL COMPUTATION

The strongest claims should be compared with methods that establish bounds, not only with methods that train networks. Theorem 1 is an application of classical stopped verification to both upper and lower witnesses. Its role here is to handle the actual liquidation trace without supposing a smooth or boundary-regular optimal value. Theorem 2 supplies a computation-robust cover, derivative, and oracle allowance. Neither theorem claims that classical verification is absent from the literature.

Monotone, stable, consistent approximations with comparison provide convergence to viscosity solutions in the framework of Barles and Souganidis (1991). Controlled Markov-chain approximation is developed systematically by Kushner and Dupuis (2001); adaptive sparse-grid methods address high-dimensional dynamic models in Brumm and Scheidegger (2017). Such methods concern specified approximation schemes and their hypotheses. The finite telescoping argument in the preserved manuscript names the operator discrepancies needed for a continuous limit; an observed mesh difference does not supply them. The original-payoff certificates instead evaluate continuous generators or exact Gaussian moments directly, with separately handled analytic and arithmetic remainders. They do not assert a convergence rate for the truncated killed-Euler scheme.

Primal-dual numerical control methods also pair feasible decisions with bounds. Brown, Smith, and Sun (2010) relax nonanticipativity with penalties and establish weak and strong optimality for stochastic dynamic programs. Brown and Smith (2011) use related bounds in portfolio optimization with transaction costs. Our no-adjustment construction is a market-deflator witness with a bounded source and explicit stopped localization; it is not a claim that policy-plus-upper-bound analysis is new. Lasserre, Henrion, Prieur, and Trélat (2008) develop occupation-measure moment relaxations for polynomial nonlinear optimal control under their stated assumptions. They do not invoke their deterministic convergence theorem for the present logarithmic, stochastic, controlled-exit problem. The explicit source bounds, interval polynomial conversions, Gaussian moments, and trace inequalities are verified here instead.

Proposition 2 and Theorem 4 identify the model-specific constructions responsible for the economic bounds. The former couples a fitted generator witness to a small localization potential; the latter cancels every portfolio term before computing a bounded-source expectation. Theorem 5 exploits rank-one action coupling and strong convexity of the remaining scalar problems. It is a structured convex-duality and branch-and-bound construction, not a generic replacement for global nonconvex optimization. Proposition 3 preserves the nonquadratic terminal payoff in a computable relaxation of the exact comparison problem.

Neural policy iteration and neural PDE solvers are important computational antecedents. Gata and Mijatović (2015) analyze continuous-time policy improvement. Kim, Cho, Kim, and Kim (2026) separate iteration, policy-network, and residual errors in entropy-regularized

on and closed-loop performance. Lee and Kim (2025) combine policy iteration and learning. Cai, Fang, and Zhou (2025) impose martingale conditions without an explicit terminal infimum. NBO’s raw training comparison is with a pinned implementation of the method, not a locally recreated approximation described as author code. Our certificates audit a policy produced by another solver. This portability is a property of independent verification, while the training results remain evidence about the specifically recorded NBO and CartNet adapters.

Extensive utility, temporal-self equilibria, and games require their own evaluation operators and optimality notions. Their complete equations, assumptions, finite gain bounds, failures, and numerical results are preserved in the supplement. No new large-scale recursive or game-theoretic result is inferred from the additive-utility certificates above. The stochastic-trace bias and local-optimizer counterexamples are preserved for the same reason: they identify conditions that a neural training loss cannot replace.

9. CONCLUSION

Our computations establish a positive minimum value of access to preference adjustment in a continuous economy. They do so by combining a feasible learned policy with an independently certified upper bound on the restricted optimum, rather than by assuming either rule is optimal. The restricted economy is enclosed within .00905 at the central state. The flexible policy’s evaluation is much more accurate than its present .10634 regret bound at the full .01 flexible-policy objective remains a distinct numerical target. Our methodological results give operational ways to improve and audit these calculations: Bernstein witnesses, stopped Gaussian-moment evaluation, market-deflator localization, neural derivatives, and a non-enumerative continuous-action oracle. The coupled computation problem also has a rigorous absolute lower bound and a frozen multi-budget, multi-holdout record. Together these results sharpen the connection between neural computation and economic conclusions while leaving every inherited derivation, experiment, and resolved diagnostic available for further review.

APPENDIX A: VERIFICATION AND COMPUTATIONAL ERROR

A.1. Selection, stopped comparison, and the neural bridge

PROOF OF PROPOSITION 1: For a fixed jet the three controls enter additively. The consumption objective has derivative $c^{-u} - v_x$ and second derivative $-uc^{-u-1} < 0$. If $v_x \leq 0$ it is increasing; otherwise its unconstrained maximizer is $v_x^{-1/u}$. The adjustment objective is $\theta^2/2$, which is strictly concave for $k > 0$. The portfolio objective is $h_1p + h_2p^2/2$. A nonnegative h_2 gives the projected vertex. A nonnegative h_2 gives a maximum at an endpoint, while the linear and constant cases. Compactness and the stated tie rule give a feasible selection. These are control-space statements at a fixed jet, not parameter-space statements. *Q.E.D.*

PROOF OF THEOREM 1: Localize the process on interior compact sets and apply discounted Bellman’s formula to an upper witness on each time slab. The discounted terminal witness plus expected reward equals the initial witness, the expected integral of $R_a U$, and the expected discounted forward jumps. The latter sum is nonpositive. Boundary comparison gives b_U . Nonnegativity of discounting and the deterministic bound $\tau \leq T$ therefore imply

$$J_k^a(t, s) \leq U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

ols proves the optimal-value bound; attainment is unnecessary. Apply the same identity to L the implemented policy. Its forward jumps are nonnegative, $g \geq L - b_L$, and $R_\pi L \geq -\epsilon_L$, ng the lower bound. Subtraction proves the regret statement. The process is stopped at the omic exit, there is no use of an unproved interior trace of the optimal value. *Q.E.D.*

ROOF OF THEOREM 2: The cover and representative intervals imply $|R_\pi v| \leq e$ and $0 \leq q$ everywhere. Hence $\sup_a R_a v = R_\pi v + \Gamma \leq e + q$ and $R_\pi v \geq -e$. Apply Theorem 1 with $L = v$, $b_U = b_L = b$, $\epsilon_U = e + q$, and $\epsilon_L = e$. A derivative bound with respect to the chosen norm gives the two moduli by integration along segments, or along a specified path inside cell. In a nonconvex cell, a verified path-length radius must replace the Euclidean radius. The shing-allowance assertion follows directly from the bound and does not require a compact of network parameters. *Q.E.D.*

he finite counterpart follows from positivity of the continuation operators: propagate the uation error with one factor δ , propagate the optimality error with one factor δ , and add the idicate gain. Backward induction from terminal error b gives (8). The full state-dependent ction, correction rule, and recursive-operator versions are reproduced in the preservation lement. Unlike this finite identity, a continuous interpretation additionally needs either ied operator discrepancy terms or an independent continuous verification route. The present inal-payoff calculation uses the latter.

A.2. Arithmetic convention

he mathematical constants specifying the economy are exact decimals. Stored network ghts, policy constants, and fitted coefficients are interpreted as exact binary64 numbers. stant formation and polynomial coefficient conversion use 70-digit interval arithmetic; ar optimization bounds use at least 50 digits. Vectorized interval addition, subtraction, iplication, division by a strictly positive interval, and pairwise summation round both oints outwards with the adjacent binary64 number. An interval square may be enclosed by ary interval multiplication; this is conservative even when its lower endpoint is negative. ranscendental library approximation is treated as a rigorous interval merely because its ut was subsequently rounded once. Exponentials in vectorized routines are replaced by or polynomials with analytic remainders; interval logarithms, trigonometric constants, and re roots are formed in the multiprecision layer.

ll derivative and residual claims concern the mathematical function represented by the stored stants, not the real-time floating-point implementation of an arbitrary neural policy. The inal time controls are themselves stored constants, so their continuous implementation is mbiguous. External-network rollouts are explicitly floating-point performance estimates. distinction is retained in every numerical table.

APPENDIX B: ORIGINAL-ECONOMY FITTED WITNESSES

B.1. Global fallback and localized upper bound

or $G = -.02(u - 2)^2 + .1 \log x$, consumption and portfolio selection give $c = .8$ and $p = .8$. adjustment gain is bounded above by $.000512/k \leq .001024$ for $k \geq .5$. The preference part $-\rho G$ is at most $.000512$. Consumption utility is maximized at the upper endpoint $u = 2.8$

izing $-.08/x - .004 \log x$ over $[.5, 2]$ gives $-.04 - .004 \log 2$. Consequently

$$G \leq \frac{.8^{-1.8}}{-1.8} + .000512 + .001024 + .00195 + .00352 - .04 - .004 \log 2 < -.8659.$$

NBO: R10 PRESERVATION SUPPLEMENT

17

$\geq g$. Stopped verification gives $V_k \leq G$ globally on the original domain, including as a continuation bound at an artificial verification face.

OF PROPOSITION 2: The derivatives of $G + F$ in x coincide with those of G . Thus the c and p selection applies. The residual after maximizing over wealth is precisely (13). $8h \exp\{\lambda(u - u_0) + \kappa h\}$, where $h = 1 - t$,

$$(\partial_t + \mathcal{L}^a - \rho)z = 8e^{\lambda(u - u_0) + \kappa h} \{-1 + h(-\kappa + \lambda\theta + .00125\lambda^2 - \rho)\}.$$

ices $(|\lambda|, \kappa) = (120, 42), (160, 64)$ satisfy $\kappa = .2|\lambda| + .00125\lambda^2$. Both terms therefore are positive killed generators for every admissible control. Moreover $(\partial_t - \rho)C_\rho(1 - t) = 0$ follows that U has nonpositive residual inside the verification domain.

$u = 1$, $F = Z = C_\rho = 0$, and $U = G$. At an actual wealth exit, $F \geq -8h$ implies $U \geq G - 8h$. At either artificial preference face, one summand of Z is at least $8h$, so $U \geq G$. Stop first at an economic exit or an artificial face. If the artificial face occurs first, the remaining original payoff is bounded by G under every admissible continuation, by the global fallback property. Concatenated stopped verification gives the asserted bound without modifying the dynamics or opportunity set. Q.E.D.

B.2. Polynomial source enclosure and refinement

$u - 1 = A + Bz$, where $A = 1.05$, $B = .55$, and $|z| \leq 1$. Let $D = (A^2 - B^2)^{1/2}$ and $(A + D) < 1$. The uniformly convergent expansion

$$\frac{1}{A + Bz} = \frac{1}{D} \left[1 + 2 \sum_{n=1}^{\infty} (-q)^n T_n(z) \right]$$

approximation error at degree 16 bounded by $2q^{17}/\{D(1 - q)\}$. If $L = -\log(.8)$, approximate the remainder by e^{LBz} times the degree-eight Taylor polynomial of e^{LBz} . Its remainder is at most $(LB)^9/9!$. Multiplying the two enclosures, with the relevant sup-norm bounds, gives a uniform error below 2.117×10^{-9} for $-.8^{-(u-1)}/(u - 1)$. This error is added to the upper bound, not fitted away.

The I_k branches are $-.2q - .02k$ for $q \leq -.2k$, $q^2/(2k)$ for $|q| \leq .2k$, and $.2q - .02k$ for $q \geq .2k$. On each box, interval Bernstein coefficients enclose q and every potentially applicable polynomial. A box touching a switching threshold includes both adjoining branches. Comparison uses outward threshold intervals; binary64 representation of $.2k$ cannot exclude a branch. A tensor polynomial lies between its minimum and maximum Bernstein coefficients. The De Casteljau subdivision preserves this enclosure and is repeated on boxes whose upper bound exceeds the prescribed ceiling. The independent check of $F/h \geq -8$ uses the same null property.

A fixed polynomial whose true residual lies strictly below a ceiling, Bernstein refinement verifies that ceiling, up to the explicitly accounted source and arithmetic allowances. It is not a certification statement about a fixed candidate with slack. It is not a theorem that fitting, a wealth-independent ansatz, or arbitrary Adam training produces approximants

APPENDIX C: INDEPENDENT EVALUATION OF THE LEARNED TIME POLICY

the projected policy has constant c_j, θ_j on each of sixteen slabs, $p = 0$, and $0 \leq \theta_j \leq .2$. Training chooses these constants, but the subsequent evaluator reads only the frozen policy. The exact wealth formula in the main text and the outward budget check show $x_1 > .5$. Since $x_0 = 1.25$ and $x_0 = 1.25$, wealth is decreasing throughout the horizon and never reaches the lower boundary. Neither Brownian correlation nor a wealth discretization affects this conclusion on $p = 0$.

Without preference stopping, $u_t = 2 + \mu(t) + .05W_t^u$, with $\mu(t) = \int_0^t \theta_r dr \in [0, .2]$. An upper-boundary exit requires $W_t^u \geq 12$ at some $t \leq 1$; a lower exit requires $W_t^u \leq -16$ and is contained in $\{\sup_{t \leq 1} |W_t^u| \geq 12\}$. The exponential martingale maximal inequality on each slab gives

$$p_{\text{exit}} \leq 2e^{-72} < 1.077 \times 10^{-31}.$$

The original preference diffusion is still stopped at exact first exit. Its rare-exit correction is not zero.

At $r = u - 2$ and $L_j = -\log c_j$. Before exit, $|r| \leq .8$ and utility is $-e^{L_j} e^{L_j r} / (1 + r)$. Replace the reciprocal by $\sum_{m=0}^{80} (-r)^m$ and the exponential by its degree-ten Taylor polynomial. The metric remainder is at most $.8^{81} / .2$; the exponential remainder is at most $e^{.8L_j} (.8L_j)^{11} / 11!$. The product bound, including the reciprocal and exponential suprema, yields a uniform error for degree-90 polynomial P_j . This bound holds on the whole original preference interval, not just at training quadrature nodes.

For a Gaussian random variable with mean μ and variance v , its raw moments satisfy $M_0 = 1$, $M_1 = \mu$, and

$$M_n = \mu M_{n-1} + (n-1)v M_{n-2}.$$

Since $\mathbb{E}P_j(u_t - 2)$ is evaluated by this recurrence with $v = .0025t$, and the terminal quadratic preference settlement has an exact Gaussian expectation. On each time rectangle the intervals of μ, v , discounting, and the polynomial coefficients enclose the entire integrand. Outward bounds of interval range times interval length therefore enclose its integral. These ranges are quadrature-error estimates inferred from two discretizations. The three resolutions are independent enclosures of the same integral.

To compare this free-Gaussian polynomial calculation with the stopped economic payoff, let E be the event of preference exit. Off E the two paths agree through T . On E , the omitted polynomial tail is bounded using

$$\mathbb{E}[\mathbf{1}_E |P_j(u_t - 2)|] \leq \sqrt{p_{\text{exit}}} \|P_j(u_t - 2)\|_2, \quad \|P_j\|_2 \leq \sum_m |b_{jm}| \sqrt{M_{2m}}.$$

Gaussian moments are bounded uniformly by evaluating their nonnegative coefficient formulas at $\mu = .2$ and $v = .0025$. The same argument bounds the free terminal settlement, and the actual exit settlement has magnitude below 8.1. Discounted adjustment cost omitted after exit is at most $.02k \mathbf{1}_E$. One valid total stopping allowance is thus

$$\sqrt{p_{\text{exit}}} \left\{ \max_j \|P_j\|_2 + \|G(u_1, x_1)\|_2 \right\} + p_{\text{exit}} (.02k + 8.1).$$

the stopping allowance is below 5.388×10^{-16} ; the utility approximation allowance is 1.151×10^{-7} before multiplying by $C_\rho(1)$. The larger remaining width comes from the y enclosed time integration, not from ignoring a boundary event.

The adjustment budget is exactly $\sum_j (\theta_j^2/2) \text{NBG}_{j/16}^{(j+1)/16}$. To preserve stopping time, reduce this by at most $.02p_{\text{exit}}$. Its interval, and the strictly positive wealth-feasibility margin, are at full precision. Tight budget intervals describe these deployed policies, not optimal-identification.

APPENDIX D: ECONOMIC COMPARISON AND THE RESTRICTED DUAL

D.1. Cost ordering and value enclosures

OF THEOREM 3: For each policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing because $B(\pi) \geq 0$. The supremum is convex and nonincreasing. The two approximate y inequalities give

$$A_1 - k_1 B_1 \geq A_2 - k_1 B_2 - \varepsilon_1, \quad A_2 - k_2 B_2 \geq A_1 - k_2 B_1 - \varepsilon_2.$$

and dividing by $k_2 - k_1 > 0$ gives (17). Zero allowances give budget ordering for optimizers at distinct prices. If π_k is optimal at k , feasibility at k_+ implies $V(k_+) \geq (k_+ - k)B(\pi_k)$; feasibility at k_- implies $V(k_-) \geq V(k) + (k - k_-)B(\pi_k)$. Substitute appropriate value-enclosure endpoints and intersect with the primitive budget range to obtain *Q.E.D.*

access lower bound follows from $V_k \geq J_k$ and $V_R \leq \bar{V}_R$. Its nonnegativity follows because the restricted policy set is contained in the flexible policy set. Likewise $V_k - V_R \leq \bar{V}_R$. These inequalities compare two optimal values even though the feasible policies giving the lower bounds need not be optimal.

D.2. Cancellation of the portfolio and stopped localization

OF THEOREM 4: For Y as specified, quadratic covariation gives

$$d(Y_t x_t) = (.04Y_t x_t - Y_t c_t)dt + \text{local martingale}.$$

Yx portfolio drift cancels the $-.06pYx$ covariation term for every adaptive p . Including the discount term, and the preference quadratic yields

$$_{,p}W_0 = \frac{c^{1-u}}{1-u} - yc + .01y - .00005 - .004 \log(.5) + .0008(u-2)^2 \leq S(u, y).$$

no W_0 mixed u - x or u - y derivative. Correlation is retained in the transition law of u used to evaluate the source potential.

source lower bound use the feasible consumption $.35$: its utility on $u \in [1.5, 2.6]$ is below by $-2/\sqrt{.35}$, while $-.35y \geq -4.2$. The remaining terms sum to a nonnegative on this rectangle. Thus $S > -7.581 > -8$. For an upper bound discard $-yc$, maximize t $c = .8$, $u = 2.6$, use $.01y \leq .12$, and bound the remaining terms by $.0033$. This gives $.698 < 0$. The clipped extension is bounded in $[-8, 0]$, so $-8h \leq F_R \leq 0$.

discounted Feynman–Kac potential satisfies the conditional expectation identity for pricing source integral. Consequently $e^{-\rho r} F_R(h - r, U_r, Y_r) + \int_0^r e^{-\rho s} \tilde{S}(U_s, Y_s) ds$ is

source is bounded. Inside the rectangle $\tilde{S} = S$, so adding the identity to discounted Itô localization for W_0 cancels its residual majorant. This argument does not require an assumed globally smooth optimal value, or differentiability of the clipped source at its outer faces. Reference localization with $\theta = 0$ uses $\kappa = .00125\lambda^2$, giving 50 and 72 for $|\lambda| = 200$ and $|\lambda| = 22$ respectively. For $l = \log Y$, $dl = -.025dt - .3dZ^x$. Each exponential localization term with $|\lambda| = 22$ is nonpositive killed generator because $22.33 = .025(22) + .045(22)^2$ dominates either sign of the log drift. As before, the factor $8h$ supplies at least $8h$ at its associated artificial face. For $x \in [.5, 2]$ and $y \geq .2$,

$$W_0 - G = y(x - .5) + .1 \log(.5/x) \geq 0 :$$

expression is zero at $.5$ and has derivative $y - .1/x \geq 0$. At an actual wealth exit the source is at least $-8h$, so $W_0 + F_R + Z_R \geq g$. At a terminal time it is at least G . At any artificial y face a localization term pays for the entire possible source deficit, giving continuation value at least G . Actual preference exit cannot precede these inner preference faces. Invoke global fallback $V_R \leq G$ after an artificial exit and apply the stopped identity before it. This verifies (25) over all original adaptive consumption and portfolio controls. The value of Y_0 is an auxiliary witness parameter, not an imposed wealth price or policy restriction. *Q.E.D.*

D.3. Validated evaluation of the restricted witness

Let z_1, z_2 be independent standard Gaussians and put $r = v^2$. At the central state,

$$u = 2 + .05v\{-.25z_1 + \sqrt{.9375}z_2\}, \quad l = \log(1.55) - .025v^2 - .3vz_1.$$

Source expectation is integrated with weight $2ve^{-.04v^2}\varphi(z_1)\varphi(z_2)$ on $v \in [0, 1]$. The box $z_2 \in [-6, 6]$ lies strictly inside the stated u, y verification rectangle under this transformation. Omission probability is at most $4e^{-18}$. Since the extended source lies in $[-8, 0]$, omission contributes an interval $[-8C_\rho(1)4e^{-18}, 0]$, whose width is less than 4.778×10^{-7} . The initial localization addition is less than 2.200×10^{-9} .

For completeness, the Hessian used by the evaluator is explicit. Let $c = \text{proj}_{[.05, .8]} e^{-l/u}$, $-\log c, y = e^l$, and let χ be one for an interior consumption solution and zero for a strictly constrained solution. Across a switching box use $\chi \in [0, 1]$. With $Q = c^{1-u}/(1-u)$,

$$Q_u = \frac{e^{L(u-1)}\{1 - L(u-1)\}}{(u-1)^2}, \quad Q_{uu} = -\frac{e^{L(u-1)}[\{L(u-1) - 1\}^2 + 1]}{(u-1)^3},$$

$$S_u = Q_u + .0016(u-2), \quad S_l = -yc + .01y,$$

$$S_{uu} = Q_{uu} + \chi yc L^2/u + .0016, \quad S_{ul} = -\chi yc L/u,$$

$$S_{ll} = -yc + \chi yc/u + .01y.$$

The envelope is continuously differentiable across clipping surfaces; its gradient is locally smooth and these piecewise second derivatives give a valid integral Taylor bound across such a face. Differentiate the displayed (u, l) transformation by the chain rule to obtain a Hessian in (z_1, z_2) . Every intermediate expression is enclosed on the entire box. Bounds on the absolute Hessian entries give the usual one-half diagonal and full mixed-term Taylor remainder.

The constant and linear Taylor terms are integrated exactly for their polynomial approximation of the weight. Gaussian weighted moments through order two use 111 terms of $e^{-z^2/2}$ and

ir analytic remainder. Source exponentials are evaluated with degree-40 Taylor polynomials on arguments in $[-3, 3]$, with error at most $e^{3941}/41!$. All antiderivatives, remainders, additions, and sums are rounded outward. The two diagonal second moments are non-negative half-linear valid Mixed moments; intersecting their enclosures with the non-negative half-linear valid Mixed moments derivatively bounded by cell-radius products times probability mass. Successive box counts are 16,384, 131,072, and 1,048,576. Their intervals enclose the initial *dual witness*. Only an upper endpoint is used to bound the restricted optimum. Using the final upper endpoint with the independently certified restricted-policy lower true optimal-value interval of width less than .009047. The distinct width below .000320 is integration of the witness, not the primal–dual gap.

APPENDIX E: CONTINUOUS ACTION AND ABSOLUTE-VALUE CERTIFICATION

E.1. The scalar global action bound and interval neural jets

OF THEOREM 5: For any action with $\sum_i a_i = z$ and any real λ ,

$$\sum_i f_i(a_i) = \sum_i \{f_i(a_i) - \lambda a_i\} + \lambda z \geq D(\lambda) + \lambda z.$$

and derivative of $.3 \sin z$ is bounded below by $-.3$. For $z = z_0 + e$, $|e| \leq r$, its quadratic minorant gives

$$\lambda z + .3 \sin z \geq \lambda z_0 + .3 \sin z_0 + (\lambda + .3 \cos z_0)e - .15e^2,$$

implies (30). Each scalar f_i has second derivative $(2 + .2 \cos 2s)/d \geq 1.8/d = m$. Thus λs is m -strongly convex. Its tangent quadratic at any feasible s_0 gives a lower bound $q'(s_0)^2/(2m)$. At an endpoint, only a feasible descent direction is relevant: an outward-minimization direction contributes zero, while an inward-pointing direction contributes negative magnitude. This follows either by minimizing the tangent quadratic over the or directly from the sign of the linear term. The stated lower bound remains valid if the minimizer lies beyond the opposite endpoint, because relaxing the displacement can over it.

sum of these scalar lower bounds is no larger than $D(\lambda)$. A feasible action gives a upper bound by direct interval evaluation. The scalar interval $[-d, d]$ covers every action sum. Subdivision, pruning only a box whose lower bound exceeds a feasible bound, and taking the minimum surviving lower bound preserve a global enclosure. Root effects efficiency but not validity. Finally $|H_p(a) - H_{\hat{p}}(a)| \leq 2 \sum_i \delta_i$ on the action box. candidate upper and true minimum lower can each worsen by at most this amount, the $4 \sum_i \delta_i$ gap correction. *Q.E.D.*

an affine layer with stored weights, interval automatic differentiation propagates both ons and spatial derivatives. A tanh layer is evaluated as $(e^{2z} - 1)/(e^{2z} + 1)$ with a positive denominator and derivative $1 - \tanh^2 z$, all in multiprecision intervals. The embedding is differentiated as its mathematical formula, including the exact $\sqrt{d}$, not as usly rounded autograd constant. This gives a componentwise interval for the derivative presented network. Subtracting the stored floating-point gradient gives the δ_i . The two tests use the inherited eight- and sixteen-dimensional learned checkpoints and the same action certificates. The summed errors are below 4.584×10^{-17} and 7.877×10^{-17} vely. These are local jet checks, not whole-state residual bounds.

PROOF OF PROPOSITION 3: For $|a| \leq 1$, $\sin(a)/a \geq \sin 1$, with the continuous value at 1. Consequently $a^2 + .1 \sin^2 a \geq r_0 a^2$, where $r_0 = 1 + .1 \sin^2 1$. Also $.3 \sin(\sum_i a_i) \geq -.3$. Replacing running cost by $r_0 \|a\|^2/d - .3$ decreases every original control's cost; allowing all $\mathbb{R}^d$ decreases the infimum further. The original terminal payoff q_d is unchanged. The relaxed HJB equation is

$$v_t + \frac{\sigma^2}{2} \Delta v - \frac{d}{r_0} |\nabla v|^2 - .3 = 0, \quad v(1, x) = q_d(x), \quad \sigma^2 = .32.$$

$\lambda = \sigma^2 r_0 / (2d)$ and use the logarithmic heat representation. At $X_0 = 0$ it gives

$$v(0, 0) = -.3 - \lambda \log \mathbb{E} \exp\{-q_d(\sigma Z)/\lambda\}, \quad Z \sim N(0, I_d).$$

Bounded cosine and coercive quadratic terminal terms give finite positive expectations. The Gaussian heat solution is smooth before the terminal date and has at most quadratic growth. Standard localization of its verification identity, or truncation followed by monotone Gaussian integration, validates the relaxed lower bound without assuming a smooth solution of the original unded-control problem.

Complete the square for the quadratic terminal component. Its Gaussian normalization contributes

$$\frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3.$$

Under the tilted Gaussian law the scalar sum divided by $\sqrt{d}$ has mean $m = 2\sqrt{d}/(r_0 + 4)$ and variance $s^2 = \sigma^2 r_0 / (r_0 + 4)$. Orthogonal coordinates disappear, leaving

$$v(0, 0) = \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda \log \mathbb{E} \exp\{-(.2/\lambda) \cos(m + sZ_1)\}.$$

This is the representation in the proposition. An upper enclosure for the positive scalar expectation gives a lower enclosure for v , because $-\lambda \log$ is decreasing. That lower endpoint is before a lower bound for the original value. *Q.E.D.*

For the scalar integral let $\eta = .2/\lambda$ and $R = \lceil \sqrt{2(\eta + 50)} \rceil$. The omitted Gaussian contribution is no larger than $2e^{\eta - R^2/2}$. Interval ranges of the positive integrand times cell lengths close the integral on $[-R, R]$. The successive 1,024, 4,096, and 16,384-cell calculations add analytic tail allowance and then apply the decreasing logarithm with outward rounding. Error widths measure integration uncertainty in the relaxed value. Relaxing the action bound under the running sine can leave a large structural gap even when this quadrature width is small.

APPENDIX F: REPLICATION AND SCOPE LEDGER

The central economic certificate is independent of the inherited killed-Euler discretization. The restricted upper has a verified continuous-generator residual, an explicit source-polynomial remainder, a stopped-trace inequality, and exponentially bounded artificial-face continuation. The restricted lower has exact deterministic wealth, interval Gaussian polynomial moments, an explicit utility approximation remainder, outward time integration, and an exact-exit probability section. The restricted upper cancels continuous portfolio controls algebraically and evaluates

external experiment has a different status. Each declared sample-and-hold policy is a continuous control in the stated additive-noise dynamics. Evaluation uncertainties are for holding slab, but the stored path estimates use floating-point network evaluations and a Monte Carlo sample. The rigorous lower bound supplies an absolute reference; subtracting a sampled mean does not create an outward policy-regret certificate. Paired path errors, variation, completed-update clock overshoot, setup, and warmup are recorded separately. A sign test concerns the frozen joint training-and-audit pipeline under independent laws. It does not certify an underlying PDE or imply that the six panels represent six different problem families.

A new table is generated from raw JSON or path records; display endpoints are rounded conservatively in direction. The frozen external protocol, all tuning trials, all 72 holdout pairs, endpoints, and all path arrays are retained. The validation program checks coverage of the panels, matching initialization hashes, pinned author revision, finite outputs, refinement, primitive feasibility, the positive access lower bounds, and the explicitly false flexible decision flags. The complete R8, R4, and delivered R5 textual and numerical material is in the preservation supplement, and earlier source files remain byte-for-byte unchanged. A failed pilot nor a historical manufactured-payoff test is used as new original-economy evidence.

REFERENCES

- and P. E. Souganidis (1991): “Convergence of Approximation Schemes for Fully Nonlinear Second Order Problems,” *Asymptotic Analysis*, 4, 271–283. [13]
- B., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58(4, Part 1), 785–801. DOI: 10.1287/opre.1090.0796. [13]
- B., and J. E. Smith (2011): “Dynamic Portfolio Optimization with Transaction Costs: Heuristics and Dual Bounds,” *Management Science*, 57, 1752–1770. DOI: 10.1287/mnsc.1110.1377. [13]
- , and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85, 1575–1612. [13]
- Fang, and T. Zhou (2025): “SOC-MartNet: A Martingale Neural Network for the Hamilton–Jacobi–Bellman Problem without Explicit $\inf_{u \in U} H$ in Stochastic Optimal Controls,” *SIAM Journal on Scientific Computing*, 47(4), 1581–1619. DOI: 10.1137/24M1681033. [14]
- Fang, and T. Zhou (2025): “SOC-MartNet: A Martingale Neural Network for the Hamilton–Jacobi–Bellman Problem without Explicit $\inf H$ in Stochastic Optimal Controls,” arXiv:2405.03169, version 4, March 15. Historical citation retained for the preservation supplement. []
- and L. G. Epstein (1992): “Stochastic Differential Utility,” *Econometrica*, 60, 353–394. []
- G., and S. E. Zin (1989): “Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Returns: A Theoretical Framework,” *Econometrica*, 57, 937–969. []
- and W. E (2016): “Deep Learning Approximation for Stochastic Control Problems,” arXiv:1611.07422. []
- Jentzen, and W. E (2018): “Solving High-Dimensional Partial Differential Equations Using Deep Learning,” *Proceedings of the National Academy of Sciences*, 115, 8505–8510. []
- , and A. Mijatović (2015): “On the Policy Improvement Algorithm in Continuous Time,” arXiv:1509.09041. []
- ... (1998): *Numerical Methods in Economics*. MIT Press. []
- Cho, M. Kim, and Y. Kim (2026): “Physics-Informed Approach for Exploratory Hamilton–Jacobi–Bellman Problems via Policy Iterations,” *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(27), 22609–22616. DOI: 10.1609/aaai.v40i27.39421. [13]
- M. Kim, Y. Kim, and N. Cho (2026): “Physics-Informed Policy Iteration for High-Dimensional Hamilton–Jacobi–Bellman Equations: Interior Error Bounds without Boundary Data,” arXiv:2508.01718, version 2, August 9.

formed Approach,” arXiv:2508.01718, version 1. Historical citation retained for the preservation supplement; see updated 2026 entry for the current title and author order. []

ner, H. J., and P. G. Dupuis (2001): *Numerical Methods for Stochastic Control Problems in Continuous Time*, second edition, Springer. DOI: 10.1007/978-1-4613-0007-6. [13]

erre, J. B., D. Henrion, C. Prieur, and E. Trélat (2008): “Nonlinear Optimal Control via Occupation Measures and H-Relaxations,” *SIAM Journal on Control and Optimization*, 47, 1643–1666. DOI: 10.1137/070685051. [13]

J. Y., and Y. Kim (2025): “Hamilton-Jacobi Based Policy-Iteration via Deep Operator Learning,” *Neurocomputing*, 5, 130515. DOI: 10.1016/j.neucom.2025.130515. [14]

J. Y., and Y. Kim (2024): “Hamilton-Jacobi Based Policy-Iteration via Deep Operator Learning,” arXiv:2406.10920. Historical preprint citation retained for the preservation supplement. []

on, R. C. (1969): “Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case,” *Review of Economics and Statistics*, 51, 247–257. []

Q. (2025): “Neural Foundations of Endogenous Risk Aversion,” unpublished manuscript, Peking University; motivating manuscript identified in the retained NBO source. []

PART II: COMPLETED R9 SUPPLEMENT, PRESERVED IN FULL

This supplement preserves the complete R8 manuscript and proofs, the complete R4 exposition and derivations, and all six delivered R5 numerical tables without altering their source text. It also specifies the R9 replication and evidence map. Recursive utility, temporal selves, dynamic games, Merton benchmarks, stochastic-trace identities, finite safeguards, manufactured-payoff tests, and unsuccessful numerical attempts remain available for review. The new original-economy welfare bounds, constructive certificates, and frozen external comparison are developed in the R9 main paper and its mathematical appendix.

PART I: READING AND REPLICATION MAP

The current manuscript is `ECTA_R9.tex`; its Appendix proves the new results. The accompanying response addresses the September 22, 2026 R8 numerical-methods report at commit `9c3c`. The new work is isolated on the R9 constructive-certification branch. Earlier main review branches have not been advanced by this revision.

The R9 replication directory contains the original-payoff actor and moment evaluator, fitted- ϵ -construction and Bernstein verifier, restricted market-deflator verifier, scalar global action verifier, interval neural-jet evaluator, absolute-value relaxation, and frozen external-comparison verifier. Numerical tables are generated from their retained JSON records. Every new trained neural network and all 4,096-path samples for each of the 72 holdout pairs are retained. Source preservation hashes, exact commands, scope checks, and failed-pilot status are in the R9 replication and validation records.

The following material is historical but preserved in full. References to a preceding revision, its experiment, its current target, or its conclusions refer to that historical record. In particular, the R8 manufactured-payoff result is not evidence for original-payoff precision, and its six-seed neural comparison is superseded for current inference by the frozen R9 holdout. Neither page is deleted or silently rewritten. Historical finite safeguards remain distinct from the new continuous-action oracle. The R4 recursive, temporal-self, and game derivations retain their own assumptions and equilibrium concepts; no new large-scale extension result is inferred from the finite-control computations.

accurately must an economic decision rule be computed before a welfare comparison is possible? This question is especially consequential when interventions make changes not only to the economic environment but also a preference state. A small error in a fitted value function need not imply a small policy loss, and a numerically stable policy surface need not imply an adjustment-cost effect from an approximation error. The difficulty is compounded when the economic boundary is an actual termination event rather than an artificial edge of a domain.

This paper develops Neural Bellman Operators (NBO) as a model-based framework for policy evaluation, feasible policy improvement, and verification of the implemented policy. The economic application combines stochastic risk aversion, consumption, portfolio choice, and costly preference adjustment. Its stopping and settlement contract is specified as a function of the economy. The mathematical results identify three distinct objects: an exact policy-improvement sequence, its neural implementation, and an independently evaluated policy. None is trusted for another in the error bounds.

The first result provides a continuous-time verification pair. A supersolution controls the value of an admissible policy, whereas a subsolution controls the payoff of the particular policy implemented. Their difference, together with explicitly signed residual and boundary conditions, bounds regret. The argument uses smooth test functions, not smoothness of the optimal policy; therefore does not require continuity of the optimal value at an irregular controlled boundary. We give both a direct continuous-generator implementation and an operator-splitting decomposition for a discretized implementation. The latter separates time, action, interpolation, quadrature, exit, inner-solve, and arithmetic errors; a refinement difference is the sum of these bounds.

The second result connects computational accuracy to an economic comparative static. Raising the cost of preference adjustment lowers the optimal discounted adjustment budget. For policies that are only approximately optimal, the possible reversal in this ordering is bounded by the change in their regret bounds divided by the change in price. Value enclosures also bound optimal adjustment budgets through secants and bound the welfare gain from access to the adjustment technology. These statements make the required numerical precision depend on the economic problem, rather than on a common uncalibrated stopping tolerance.

Computational analysis has complementary components. We re-evaluate the retained policies, price the full finite-action safeguard, and separate time, state, and action components. We construct outward-interval verification pairs for their deployment in the original economic diffusion. Those pairs are valid but too conservative to resolve the stated economic problem in target. A second, explicitly modified-payoff experiment trains a bounded actor on the same state, control, covariance, and exit geometry and obtains an all-state continuous certificate. Finally, a coupled nonconvex control problem in eight and sixteen dimensions is solved at a fixed CPU budget with the pinned author implementation of SOC-MartNet. The solution uses actual neural controls, no analytical greedy-action labels, paired independent rollouts, and a declared six-seed design. Its outcome measures realized cost at that budget, relative to an unknown optimum.

Policy iteration, neural PDE evaluation, and their combination have important antecedents. The study by Mijatović (2015) study continuous-time policy improvement. Kim, Cho, Kim, and Kim (2026) alternate soft evaluation and improvement for entropy-regularized stochastic control. The separate iteration, policy-network, and residual errors. The updated interior analysis of Kim, Kim, and Cho (2026) treats stationary second-order equations with bounded-domain

ed-loop mismatch are directly relevant. Lee and Kim (2025) combine policy iteration with actor learning to reuse a representation across terminal data. Cai, Fang, and Zhou (2025) use stochastic control and value networks through martingale conditions without explicitly computing the Hamiltonian infimum. NBO does not claim priority for neural policy iteration or residual verification. Its contribution here is the combination of stopped economic-model verification, explicitly costed feasible correction, and error-calibrated preference-adjustment comparisons. Section 7 gives the comparison at the level of assumptions and implemented experiments.

Recursive utility, sophisticated temporal selves, and dynamic games remain part of the network. They do not share an identical evaluation equation or an identical optimality notion. We retain their aggregators, equilibrium conventions, proofs, and computational records, and we show precisely how a gain bound changes when the evaluation operator is nonlinear or when rivals are frozen. The supplementary material preserves the complete preceding R4 manuscript and technical derivations; the current manuscript integrates the R5 evidence and the new continuous verification and external-comparison results.

2. AN ECONOMY WITH COSTLY PREFERENCE ADJUSTMENT

The state is $s = (u, x)$, where u is relative risk aversion and x is financial wealth. An admissible control is a progressively measurable process $a = (c, \theta, p)$ taking values in

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

Before termination the state satisfies

$$du_t = \theta_t dt + .05 dZ_t^u, \quad (1)$$

$$dx_t = [(.02 + .06p_t)x_t - c_t]dt + .2p_t x_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

The horizon is $T = 1$, the open state domain is $D = (1.2, 2.8) \times (.5, 2)$, and the discount rate is $\rho = .04$. The process is stopped at $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. Its running payoff and termination fee are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad k > 0, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

s

$$J_k^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k(t, s) = \sup_\pi J_k^\pi(t, s). \quad (5)$$

For any k , feasible controls, state dynamics, stopping rule, and settlement do not depend on k . Bounded controls and locally Lipschitz state coefficients give a well-defined stopped state equation for the process. The feedbacks used below. Bounded payoffs on the stated domain give finite values. The cardinalization and consumption units are held fixed across comparisons. A transformation of the economy depending on the controlled state u is generally a different economy.

The early termination fee is a specified liquidation covenant, not an estimated preference parameter. At termination there is no subsequent wealth injection or continuation consumption. This distinction is essential: a constant preference diffusion cannot be confined to the stated

drift is $-.016$, so that the original consumption floor cannot support a silently viable
 ncing rectangle. We do not replace this economy by reflection.
 e upper wealth boundary, the admissible choice $p = 0$ has zero wealth diffusion and
 drift $.04 - c < 0$. Consequently, a theorem that assumes boundary regularity uniformly
 controls cannot simply be applied to this model. We neither identify the interior trace of
 the prescribed immediate-settlement value without proof nor assume a globally smooth
 value. The verification pair in Section 4 uses inequalities on the actual stopped trace
 gains meaningful in this setting.

smooth test function v , define

$$\mathcal{L}^a v = \theta v_u + [(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$\mathcal{R}_a v = v_t + \mathcal{L}^a v + \ell_k - \rho v. \quad (7)$$

rior HJB equation is $\sup_{a \in A} \mathcal{R}_a V_k = 0$ in the appropriate solution sense. All covariance
 including the mixed derivative, are retained. A negative shock correlation by itself does
 abolish the sign of the optimal portfolio response, and dividing a re-solved V_u by k does
 abolish a pointwise ordering of adjustment policies.

POSITION 1—Complete control selection for a fixed jet: *Let v have the derivatives in
 global maximizer of $\mathcal{R}_a v$ is obtained separately in the three control components. If $v_x > 0$,
 $\text{proj}_{[.05, .8]}(v_x^{-1/u})$; otherwise $c^* = .8$. Also $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. Write*

$$h_1 = .06xv_x - .0025xv_{ux}, \quad h_2 = .04x^2v_{xx}.$$

0, then $p^* = \text{proj}_{[-.5, .8]}(-h_1/h_2)$. If $h_2 \geq 0$, choose an endpoint maximizing $h_1p +$
 with a fixed tie rule.

selection rule includes nonpositive marginal wealth value, zero curvature, convex port-
 jectives, and constrained optima. It removes action discretization from a continuous-
 or audit for this particular model. It is not a closed-form optimizer of the killed-Euler,
 ated, finite-step Bellman target, and it is not a global convergence theorem for shared
 parameters. The underlying derivative rule was present in the retained derivations; the
 implementation tests every curvature branch explicitly.

3. EVALUATION, IMPROVEMENT, AND FINITE CORRECTION

fixed policy π , policy evaluation solves $R_\pi v = 0$ with the economic stopping contract.
 ment chooses a feasible control maximizing the Hamiltonian of that fixed evaluation.
 ct operators are denoted $\mathcal{E}$ and $\mathcal{I}$, and exact policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. If a
 evaluation exists, $R_{\pi_{j+1}}(\mathcal{E}\pi_j) \geq 0$. Verification then gives $J^{\pi_{j+1}} \geq J^{\pi_j}$.
 act limiting argument requires more than this monotonicity. Under compactness of
 ey class, continuity of evaluation into a precompact $C^{1,2}$ jet class, closure under im-
 ent, continuity of the chosen selector at cluster points, and comparison for the optimality
 a, each cluster-point value is optimal. These are hypotheses on the exact operators.
 not follow from successful stochastic-gradient training. Appendix B gives the limiting
 at and retains the distinctions between parameter stationarity, policy convergence, and
 nvergence.

critic v_ξ is trained against fixed-policy targets. In a continuous-generator implementation they are $R_\pi v_\xi$; in a discrete implementation they are the appropriate one-step conditional expectations. The actor improves a frozen critic. Freezing the critic's parameters does not mean freezing the derivative of its output with respect to the controlled next state. Conversely, no actor state may change the critic parameters or a rival's policy parameters through the improvement objective. Independent policy evaluation is a third operation and supplies neither optimal-action nor optimal-value training labels.

In the NDU implementation inherited from R5, the raw actor has three bounded outputs through two tanh hidden layers. It is trained without enumerating actions, but its training states are on a lattice. It is therefore a continuous-action, lattice-trained implementation, not a mesh-based experiment. The high-dimensional experiment below instead uses sampled states and Gaussian successor samples. The two implementations share the separation of evaluation and improvement; their verification costs are different.

3.2. A certificate for the declared finite economy

Let $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, where P_n^a is a positive substochastic continuation matrix and $\delta = \Delta$. Termination payoffs are included in r_n^a . Let $\hat{\pi}$ be a feasible policy and v_n a critic, with minimal discrepancy at most b . Define

$$e_n = \|v_n - \mathcal{T}_n^{\hat{\pi}} v_{n+1}\|_\infty, \quad \eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\hat{\pi}} v_{n+1}\|_\infty.$$

THEOREM 1—Finite gain accounting: *On this finite economy,*

$$\|V_{h,n}^* - V_{h,n}^{\hat{\pi}}\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

where $V_h^{\hat{\pi}}$ is independently evaluated with the exact finite terminal data, $e_j = b = 0$. A state-dependent bound is obtained by setting $B_N = 0$ and

$$B_n(s) = g_n(s) + \delta \max_a (P_n^a B_{n+1})(s), \quad g_n = \max_a \mathcal{T}_n^a V_{h,n+1}^{\hat{\pi}} - V_{h,n}^{\hat{\pi}}.$$

$$\text{then } 0 \leq V_{h,n}^* - V_{h,n}^{\hat{\pi}} \leq B_n.$$

where the maximum must range over the entire declared candidate economy. For the retained JUD audit this is the 125-point action grid together with the frozen continuous neural proposal for every state and date. The certificate does not cover all continuous controls merely because the candidate originated in a neural network. Floating-point computation of the finite envelope is recorded separately from the real-arithmetic statement of the theorem.

3.3. The safeguarded algorithm and its cost

The safeguarded algorithm first evaluates the raw policy, computes all candidate gains, and rejects states with gain above a specified threshold. It re-evaluates the corrected policy and acts only when its reported envelope meets the target. In the new re-audit the threshold is

$$\gamma = \frac{\varepsilon}{2 \sum_{n=0}^{N-1} \delta^n}. \quad (9)$$

test remains the recomputed envelope, not an optimal reference value. A stopped run
s not meet that test is a failed precision test.

POSITION 2—When corrections can become arbitrarily small, the sufficient condition for a vanishing correction share at a fixed threshold. *With nonnegative gains g_m , the share whose gain exceeds $\gamma > 0$ is at most $(M\gamma)^{-1} \sum_m g_m$. A vanishing mean gain is sufficient for a vanishing correction share at a fixed threshold. As $\gamma_j \downarrow 0$, the sufficient condition is $M_j^{-1} \sum_m g_{j,m} = o(\gamma_j)$.*

31

s a conditional mechanism, not evidence that longer training has already made the
corrections negligible. It distinguishes a useful limit statement from an unmeasured
dose claim. Every audit still enumerates the candidate actions. With N dates, S lattice
and A grid actions, the implementation uses $NS(A + 2)$ value-action evaluations per
including policy evaluation and the extra proposal, and $NS(A + 1)$ envelope-action
evaluations. Each NDU action uses four shock branches and bilinear interpolation. There is no
non-free cost claim for this safeguard.

4. VERIFICATION OF THE CONTINUOUS ECONOMY

subject implemented in continuous time must be stated before it can be certified. For the
NDU proposals, we use their feasible bilinear interpolation on the exact uniform state
at each decision interval the feedback function is frozen in time, but is evaluated at the
state continuously: $a_r = \pi_n(S_r)$. This is not a simulation that holds the initial action
at the next date. It defines a bounded, piecewise-time, Lipschitz state-feedback rule for the
Brownian dynamics and exact first-exit contract.
 $C_\rho(h) = (1 - e^{-\rho h})/\rho$, with $C_0(h) = h$.

ASSUMPTION 1—Admissible verification witnesses: *U and L are bounded, continuous
on the stopped trace, and $C^{1,2}$ in the interior of each time slab. Their derivatives permit
Itô verification and the stopped expectations are uniformly integrable. The policy π
satisfies the well-defined stopped state equation. At interior slab interfaces either the witnesses are
continuous or they satisfy $U(t^-, s) \geq U(t^+, s)$ and $L(t^-, s) \leq L(t^+, s)$.*

THEOREM 2—A one-sided continuous verification pair: *Under Assumption 1, suppose
the pair is continuous on the open domain*

$$\sup_{a \in A} R_a U \leq \epsilon_U, \quad R_\pi L \geq -\epsilon_L,$$

*where the actual terminal and exit trace $U \geq g - b_U$ and $L \leq g + b_L$, where all four constants
are nonnegative. Then*

$$V(t, s) \leq U(t, s) + b_U + C_\rho(T - t)\epsilon_U, \quad (10)$$

$$J^\pi(t, s) \geq L(t, s) - b_L - C_\rho(T - t)\epsilon_L, \quad (11)$$

$$0 \leq V - J^\pi \leq U - L + b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (12)$$

where the value need not belong to $C^{1,2}$ or be continuous at every controlled boundary point.

Two witnesses need not be neural networks, and need not have the same approximation
power. A trained network can propose either witness, but verification requires an enclosure

enclosure. The symmetric residual certificate is recovered by taking $U = L = v$, with $|R_\pi v| \leq e$, $R_a v - R_\pi v \leq \eta$, and $|v - g| \leq b$: regret is at most $2b + C_\rho(2e + \eta)$. Spatially localized errors can be handled without replacing them by a sampled mean. If $R_a U \leq \mathfrak{H}(s, t)$, choose a nonnegative potential z_U with zero-or-positive terminal/exit $z_{U,t} + \mathcal{L}^a z_U - \rho z_U \leq -q_U$ for every action. Then $U + z_U$ is an exact upper witness. The analogous potential for the fixed policy converts L into a lower witness. These computation-sensitive potentials identify a constructive route to tighten conservative global certificates; they themselves must be verified rather than estimated on favorable trajectories.

4.1. An explicit original-payoff construction

For G in (4), Proposition 1 gives $c = .8$, $p = .8$, and $\theta = -.02(u - 2)$ when $k = 2$. A 160-by-160 outward-interval partition bounds $\sup_{s,a} R_a G$ above by $-\alpha$, with $\alpha > 0$. For each frozen policy we likewise bound $R_\pi g$ below by $-\beta$ on every state cell and time slab. Consequently,

$$U = G - \alpha C_\rho(1 - t), \quad L = g - \beta C_\rho(1 - t) \quad (13)$$

valid exact-sign witnesses. Since $\alpha < 8$, $U \geq g$ on the stopped trace, while $L \leq g$. Their zero difference is

$$8 + (\beta - \alpha)C_\rho(1). \quad (14)$$

The interval program encloses logarithms, powers, products, and all bilinear-action component values. Input network outputs are frozen binary64 constants interpreted as exact real numbers; the state grid is the exact uniform rational grid. The returned bound is rounded outward again and converted to binary64. This certifies the specified real-arithmetic feedback, not every floating-point implementation of the stochastic differential equation.

TABLE I
ORIGINAL-PAYOFF CONTINUOUS NDU WITNESS BOUNDS

N	Interval cells	α	β	Regret upper bound
6	4320	0.864832	7.078878	14.091405
12	8640	0.864832	1.211230	8.339561

obtained raw seed-10 nodal policies, deployed as continuously reevaluated bilinear feedback on each time slab. The bounds cover the original continuous Brownian model and exact first exit using outward interval arithmetic. Both bounds exceed the .01 precision target.

The original-payoff bounds in Table I are numerically wide. They replace an absent continuous certificate by an executed, valid certificate, but do not establish regret below .01 or .05. In particular they are not a substitute for a sufficiently narrow welfare enclosure. The theorem and implementation make that distinction testable rather than hiding it in a missing discretization.

4.2. A complete discretization accounting route

There is a second route to continuous verification. Let $\mathcal{B}_n^*$ be the exact continuous-time maximal Bellman semigroup over one decision interval, allowing all admissible within-interval controls, and let $\mathcal{B}_n^\pi$ be the exact semigroup for the declared continuously implemented policy.

$$\|w_n^* - \mathcal{B}_n^* w_{n+1}^*\|_\infty \leq \kappa_n^*, \quad \|w_n^\pi - \mathcal{B}_n^\pi w_{n+1}^\pi\|_\infty \leq \kappa_n^\pi,$$

minimal discrepancies b^* and b^π . A telescoping sum argument yields

33

$$\|V_0 - J_0^\pi\|_\infty \leq B_h + \delta^N (b^* + b^\pi) + \sum_{n=0}^{N-1} \delta^n (\kappa_n^* + \kappa_n^\pi), \quad (15)$$

B_h bounds the lifted finite optimal–policy gap. Each κ can be enclosed by a declared chain of intermediate operators separating time freezing (including within-step feedback), restriction of state interpolation, shock quadrature, first exit and settlement timing, inner optimization, and cutoff. The sum is valid when each adjacent operator difference is bounded on the lifted continuation function. A null term is unknown, not zero. A coarse–fine solution scheme alone bounds none of the uncomputed tails. Nor does a regular-domain scheme automatically apply at an irregular controlled exit boundary. The direct generator pair assesses these scheme errors by auditing the continuous implemented policy itself; it does not that the errors of the training scheme vanish.

5. ECONOMIC COMPARISONS AT FINITE NUMERICAL ACCURACY

an initial state and suppress it from the notation. For each feasible policy write

$$A(\pi) = \mathbb{E} \left[\int_t^\tau e^{-\rho(r-t)} \frac{c_r^{1-u_r}}{1-u_r} dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(r-t)} \frac{\theta_r^2}{2} dr.$$

$$B(k) = \sup_\pi \{A(\pi) - kB(\pi)\}, \text{ and } 0 \leq B(\pi) \leq .02C_\rho(T-t).$$

LEMMA 3—Adjustment costs and approximate optimality: *If the feasible policies and the contract are independent of k , V is convex and nonincreasing. For $k_1 < k_2$, optimal policies satisfy $B(\pi_2) \leq B(\pi_1)$. More generally, if $V(k_i) - J_{k_i}^{\pi_i} \leq \varepsilon_i$, then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (16)$$

For restricted budgets satisfy $|\widehat{B}_i - B(\pi_i)| \leq \xi_i$, their possible reversal is bounded by the right side plus $\xi_1 + \xi_2$. At a differentiability point with an attained optimizer, $V'(k) = -B(\pi_k)$.

The coefficient in (16) is sharp without further restrictions on the feasible policy set. The lemma concerns discounted adjustment expenditure, not pointwise values of $\theta(t, u, x)$. The monotonicity statement is an economic theorem for the original continuous contract; the numerical experiment must still establish sufficient accuracy to quantify its magnitude.

PROPOSITION 3—Value intervals, budgets, and access to adjustment: *Suppose $L(k) \leq U(k)$ and $k_- < k < k_+$. Every attained optimal adjustment budget at k belongs to*

$$\left[\max \left\{ 0, \frac{L(k) - U(k_+)}{k_+ - k} \right\}, \min \left\{ .02C_\rho(T-t), \frac{U(k_-) - L(k)}{k - k_-} \right\} \right]. \quad (17)$$

The intervals $[J_F]$ and $[L_R, U_R]$ enclose the flexible and the restricted $\theta \equiv 0$ values, the value of access to adjustment belongs to $[L_F - U_R, U_F - L_R]$, intersected with $[0, \infty)$.

across prices separated by Δk , an error budget of order $\Delta k d_B$ is required, together with a target-evaluation enclosure. An arbitrary .05 value target need not suffice. Likewise, a positive measured flexible-minus-restricted value is not a certified welfare gain unless its lower interval point is positive. The utility units here are explicitly normalized lifetime utility; they have been converted into an empirically calibrated consumption-equivalent welfare metric.

6. NUMERICAL EVIDENCE

The experiments distinguish numerical validity, achieved precision, and comparative performance. The original-payoff NDU certificate in Table I is an outward-interval bound for a continuous diffusion. The finite gain envelopes in this section are float64 calculations for their shared candidate economies. The learned-geometry experiment has a rigorous continuous verification pair but a deliberately modified running payoff. The external comparison estimates fixed costs and sampling uncertainty; it does not know the optimal value. These distinctions are part of the interpretation of the results, not alternative names for the same error.

The complete new panel was executed from source commit 14ecbd48 in a clean Python 3.11 environment, with PyTorch 2.10.0, NumPy 2.3.5, SciPy 1.17.0, and mpmath 1.3.0. Numerical `mpmath` used one CPU thread. The ledger retains the processor description, exact versions, source hashes, external-code hashes, checkpoint weights, rollout arrays, all seed-level records, and execution logs. Table generation reads those committed records and fails on missing cells. The full integration runs are development checks, not extra independent confirmatory seeds.

6.1. Preserving a learned policy through thresholded correction

Table II re-audits two retained policies without retraining or selecting a new policy using the reference optimum. The twelve-date seed-10 policy had a raw localized envelope of .0922420. The preceding all-positive-gain safeguard replaced 54.6602 percent of interior state-date decisions and obtained a .00544535 envelope. The threshold in (9), at target .05, replaces only 4.935 percent and obtains a .00682279 envelope. At target .01, the correction share is 4.935 percent and the envelope is .00676060. Thus most of the earlier replacements were unnecessary in meeting the declared finite-economy tolerance in this sample.

TABLE II
THRESHOLDED CORRECTION OF RETAINED NDU POLICIES

N	Target	Raw bound	Final bound	Override (%)	Audits	Seconds
6	0.05	0.042805	0.042805	0.000	1	0.472
6	0.01	0.042805	0.003278	8.796	2	0.849
12	0.05	0.092242	0.006823	0.875	2	1.641
12	0.01	0.092242	0.006761	4.935	2	1.630

float64 finite-candidate envelopes, not continuous-model certificates. Seconds include all listed audits and envelope calculations on the current worker. Each audit still enumerates the full 125-action grid plus the proposal. Historical training time is not added across different environments.

This improvement addresses the role of correction rather than concealing it. The corrected policy differs from the raw rule, and every audit still enumerates 125 grid actions plus the neural proposal. For the twelve-date runs, two audits use 2,362,200 value-action evaluations and 3,600 envelope-action evaluations, each with four transition branches. The new policy is not

propagation, and re-audit time. Historical training time is retained separately because measured in a different run and environment; adding it to current audit time would not be machine end-to-end timing experiment.

ix-date policy already passes the .05 target without correction. At .01 it requires an percent correction share. These are deterministic post-review re-audits of the specified policies, not a new population estimate of raw training success. Proposition 2 explains dition under which correction shares vanish, but this two-policy exercise does not trate a training-dose convergence curve. Neither its finite envelope nor its smaller on share closes the continuous NDU error budget.

6.2. Separating time, state, and action approximation

III varies one approximation factor at a time. The time experiment fixes a 25×31 lattice points per action component; the state experiment fixes 48 dates and five action points; on experiment fixes 48 dates and the 25×31 lattice. All retain the original four-sign le and the straight-segment liquidation treatment. The latter two approximations are not dently refined in this panel.

TABLE III
ONE-FACTOR NDU APPROXIMATION DIAGNOSTICS

Factor	N	State nodes	n_a	Value	Exit prob.	h^2/Δ
time	24	25×31	5	-1.451486	0.104315	0.1067
time	48	25×31	5	-1.549787	0.193358	0.2133
time	96	25×31	5	-1.608530	0.227492	0.4267
state	48	25×31	5	-1.549787	0.193358	0.2133
state	48	37×46	5	-1.444048	0.151095	0.0948
state	48	49×61	5	-1.391646	0.102389	0.0533
action	48	25×31	3	-1.606541	0.245260	0.2133
action	48	25×31	5	-1.549787	0.193358	0.2133
action	48	25×31	9	-1.536711	0.182784	0.2133

s use $k = 2$ and evaluate $(u, x) = (2, 1.25)$. n_a is the number of nodes per action component. Four-sign e and straight-segment liquidation are unchanged; these differences do not bound continuous-model error.

time-only sequence is not evidence of convergence. Its central value moves from 19 at 24 dates to -1.60853 at 96 dates, while the exit probability rises substantially. g the spatial mesh fixed makes h^2/Δ increase, rather than shrink; the experiment ex- is interaction instead of fitting an unjustified time order. Conversely, refining the state t 48 dates moves the value from -1.54979 to -1.39165 . Action refinement improves e optimum, but the three approximation factors cannot be conflated. The operator- ent regression is close to machine precision; agreement of two implementations of the proximation is not an enclosure of its continuous-model bias.

ther panel uses 96 dates, a 49×61 lattice, and nine points in each action component, ctions. Each adjustment-cost economy is re-solved. Table IV shows the same ordering tment expenditure as Theorem 3, with an active preference margin. This is a stronger tion resolution than the earlier five-point panel, but still not a verified continuous welfare ce. The first-exit and quadrature terms in (15) remain unbounded by these calculations. ngly, neither a Richardson remainder nor a continuous policy-convergence rate is inferred s table.

	k	Value	Budget B	Exit prob.	θ_0	p_0
36	0.5	-1.416240	0.016811	0.162725	0.2000	0.3125
	2	-1.437379	0.011583	0.162286	0.2000	0.3125
	8	-1.468043	0.001888	0.158876	0.1000	0.3125

96, a 49×61 state lattice, and nine nodes per action component (729 actions). Central consumption is .8 in these Values and budgets belong to the declared numerical liquidation economy, without a continuous error enclosure.

6.3. A learned actor on the full NDU control and exit geometry

To test whether continuous verification transfers from an analytic example to an actually learned control, retain (2), the three-dimensional action box, correlated shocks, state domain, and settlement exactly. Change only the running payoff to

$$\tilde{\ell}(t, s, a) = \ell_2(s, a) - \sup_{b \in A} R_b g(t, s). \quad (18)$$

The subtracted term is independent of the chosen action, but depends on state and time. Thus it generally changes optimal behavior relative to the original-payoff economy; it is not an innocuous preference normalization. In the modified problem $\sup_a \tilde{R}_a g = 0$ and the explicit stable maximizer is $(.8, -.02(u - 2), .8)$. Itô verification proves that the modified value is g and the original stopping contract.

We train a four-parameter actor with constant bounded sigmoid consumption and portfolio outputs and an affine-tanh adjustment output. The actor maximizes the Hamiltonian of the own witness, with no optimal-action labels. Six declared seeds each receive 4,000 Adam steps learning rate .02 and 256 random states per step. This is a controlled verification experiment, not a deep-network capacity or high-dimensional scaling claim. The witness is known rather than learned; the actor and all of its achieved errors are learned.

For the frozen weights, the control gap is bounded over the entire state domain by outward interval arithmetic. The consumption gain is bounded using

$$\int_c^{.8} z^{-u} dz - .1(.8 - c)/x \leq (.8 - c)\{c^{-2.8} - .05\},$$

where the portfolio gain is the exact quadratic difference, and the adjustment gain equals $(\theta + .02(u - 2))$ at $k = 2$. A 2,048-cell interval partition in u encloses the latter. The full regret is bounded by $C_\rho(1)$ times the summed action-gap enclosure. The time discretization of a rollout and a numerical first-exit probability do not enter this certificate, because its proof is for the original continuous generator.

Every seed meets the .01 modified-problem target, with continuous regret bounds between .81053 and .00112263. Table V separately reports componentwise control errors rather than using value accuracy as policy sup-norm accuracy. This is the desired transfer test with a learned actor and the original first-exit/control geometry. It does not identify the original NDU error, and does not tighten Table I merely by sharing the state equation.

Seed	Regret bound	c error	θ error	p error	Train (s)
300	0.00081494	0.0004555	0.0000318	0.0007034	2.774
301	0.00103566	0.0005788	0.0000344	0.0007415	2.749
302	0.00081053	0.0004532	0.0000308	0.0005803	2.778
303	0.00112262	0.0006277	0.0000397	0.0005607	2.761
304	0.00112212	0.0006273	0.0000635	0.0006387	2.794
305	0.00099308	0.0005551	0.0000334	0.0007077	2.780

NBO: RLO PRESERVATION SUPPLEMENT

37

upper bounds use outward interval arithmetic and the same state, action, covariance, and first-exit geometry economic model. Only the running payoff is manufactured. Four actor parameters are trained for 4,000 steps without optimal-action labels; the critic witness is known.

6.4. Coupled nonconvex control and an external author implementation

new comparison does not use the Cole–Hopf quadratic-control model or regress to an greedy target. Starting from $X_0 = 0$ in $\mathbb{R}^d$, minimize

$$\int_0^1 f(a_t) dt + q(X_1) \Bigg], \quad dX_t = 2a_t dt + \sqrt{2}(.4)dW_t, \quad a_t \in [-1, 1]^d, \quad (19)$$

$$f(a) = d^{-1} \sum_{i=1}^d (a_i^2 + .1 \sin^2 a_i) + .3 \sin \left(\sum_i a_i \right), \quad (20)$$

$$q(x) = d^{-1} \sum_i (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right). \quad (21)$$

term couples all control components. For example, at an interior vector whose sum is the Hessian of the running cost has a negative direction along the all-ones vector for both dimensions. Hence the problem is genuinely nonconvex in the action. No closed-form solver or optimal-control labels are supplied to either method, and the quadratic exponential approximation does not eliminate this Hamiltonian.

external comparator is the unmodified `SOCMartNet.train` routine in the authors’ repository `sy-fang/MartNet`, pinned at commit `991ea8dd`. Its complete module hashes the execution manifest. The integration consists of model and network adapters and a solver that stops after a completed update at the time budget. The authors’ solver is written as a bespoke PINN comparator. The pinned implementation uses a Hessian-Hamiltonian interface; we therefore use constant diffusion in this comparison. This is a function of the implemented interface used here, not an assertion that the published SOC-MartNet framework excludes volatility control.

Both methods receive identical initial actor and critic weights within each seed. Both use a hidden layers of width 48, bounded actor outputs, a terminal-exact critic, 16 training intervals, and one CPU thread. NBO uses three evaluation updates per improvement interval, 256 sampled states, four Gaussian successors, and Adam with learning rate `.003`. SOC-MartNet retains its additional test network, RMSprop updates, and the declared martingale and Hessian losses. Its actor/value learning rate is `.003/ $\sqrt{d}$` , the test-network learning rate is `.01`, and the remaining adapter settings are fixed in the protocol. The test network has 600 outputs; all parameters and all setup costs are reported rather than excluded from the method description.

plete. Setup time is outside this primary budget and is reported separately. This matters: the n setup-plus-training time is approximately 10.69 seconds for NBO and 10.02 seconds for C-MartNet on this worker. Therefore the experiment is not described as an exactly matched second end-to-end race. NBO is evaluated first in each process, so one-time framework initialization is included in its measured setup cost. Table VII makes the distinction visible.

TABLE VI

PAIRED NONCONVEX-CONTROL COMPARISON AT A FIXED TRAINING BUDGET

d	NBO cost	SOC cost	Difference	Paired 95% interval
8	0.59567	2.31352	-1.71785	$[-2.13823, -1.29747]$
16	0.48067	2.72799	-2.24732	$[-2.51288, -1.98176]$

ence is NBO minus SOC-MartNet realized cost at 128 audit steps; negative favors NBO. Each dimension has six independent training/audit seeds, 4,096 common-random-number paths per pair, and ten seconds of training per od. Student intervals describe across-seed uncertainty, not optimality loss. Maximum paired path standard errors are 0.01098, 0.00900, respectively.

TABLE VII

MEASURED TRAINING AND SETUP RESOURCES

d	Method	Train (s)	Setup + train (s)	Outer updates
8	NBO	10.003	10.692	1536.7
8	SOC-MartNet	10.012	10.015	465.3
16	NBO	10.004	10.689	1477.3
16	SOC-MartNet	10.012	10.014	458.0

6-thread CPU means over six seeds. Identical actor/critic parameter counts within each dimension: 3,224/2,881 at 8, and 4,000/3,265 at $d = 16$. SOC-MartNet also trains its 600-output test network. Paired policy audit means are approximately .379 and .440 seconds. Outer-update definitions differ across algorithms and are not equal units of work.

At dimension eight, the paired mean NBO-minus-SOC cost is -1.71785 , with a six-seed student interval $[-2.13823, -1.29747]$. At dimension sixteen it is -2.24732 , with interval $[-2.51288, -1.98176]$. Thus the new evidence favors NBO under the specified adapter and training budget. This is a demonstrated benefit in a coupled nonconvex problem, in contrast to the retained R5 quadratic-control comparison. It is not a theorem of computational dominance, a replication of the published SOC-MartNet hardware or learning curves, or a statement that the computed policy is near optimal.

6.5. Uncertainty, mesh diagnostics, and retained failures

Each paired policy audit uses 4,096 paths with common Brownian increments for the two policies and an independent seed for each training seed. The primary outcome uses 128 time steps. A nested 64-step audit aggregates adjacent increments, so its difference from the finer audit is a matched time-mesh diagnostic. Table VI reports the maximum paired path standard error separately from the across-seed interval. The interval is based on six independent training/audit runs, with a Student approximation rather than a distribution-free coverage claim. Paths are not treated as independent training runs.

t at dimension eight, and -0.00176 and $.00951$ at dimension sixteen. These changes are smaller than the observed paired cost differences, but they are not certified bounds remaining time-discretization error. A matched-accuracy computational frontier and budget learning curves would answer additional questions not identified by the present budget design.

ior outcomes remain part of the evidence. The R5 ten-seed baseline had ten raw finite-
 esses, whereas its three twelve-date raw policies failed. Its recursive raw policies missed
 n target, and its eight/sixteen/thirty-two-dimensional quadratic-control NBO means
 rse than those of the in-house PINN-PI comparator. Those results are neither relabeled
 esses nor removed after observing the new comparison. The R4 Merton, nonlinear
 ost, boundary, recursive, temporal-self, game, and coupled linear-quadratic calculations,
 g failed pilots and incomplete runs, are reproduced in the preservation supplement. The
 ernal experiment changes the tested model and comparator; it does not retrospectively
 those historical results.

7. RELATION TO NEIGHBORING NUMERICAL METHODS

VIII separates the relevant methodological axes. Its entries summarize the cited analyses
 implemented comparison; they are not a ranking by paper title. In particular, an interior
 or theorem with unknown outer data, a stopped-control regret theorem with a contractual
 y, and a finite candidate-action gain envelope address different mathematical objects.

TABLE VIII
 RELATED METHODS AND THE OBJECTS COMPARED

	Principal mechanism or guarantee	Relation to the present evidence
ory PINN eration	Soft policy updates for entropy-regularized HJB equations; iteration, policy-network, and PDE residual errors	Neighboring policy-evaluation/improvement analysis; not the external implementation tested here
physics-policy	Stationary interior error control with residual and outer-information contributions under stated regularity conditions	Clarifies what interior collocation can control without supplying the stopped NDU boundary data
ONet	Policy iteration and reusable operator approximation across terminal data	Representation reuse is a different performance target from one trained policy at a fixed budget
rtNet	Martingale training of value and control without an explicit Hamiltonian infimum	Pinned author routine tested on coupled constrained nonconvex controls in $d = 8, 16$
this paper	Separated evaluation, improvement, and independent gain or witness verification	Continuous stopped-model theorem, explicit economic error calibration, finite correction costs, and paired author-code experiment

Kim, Cho, Kim, and Kim (2026), Kim, Kim, Kim, and Cho (2026), Lee and Kim (2025), Cai, Fang, and (2025). The new external benchmark uses constant diffusion and a ten-second training budget. Its scope does not establish a general ranking for recursive utilities, games, or higher dimensions.

an analyst choosing a method, the distinction between state dimension and action dimension is crucial. Neural state representations avoid a tensor state grid during the high-dimensional

ponent selection in Proposition 1 is specific to that economic Hamiltonian. The coupled control problem deliberately removes that separability. Likewise, none of the numerical comparisons establishes that a small sampled training loss satisfies the uniform hypotheses of continuous certificate. The contribution is a verifiable separation of these issues, with one favorable external comparison and one original-economy precision test whose conservative outcome remains visible.

8. RECURSIVE UTILITY AND STRATEGIC DECISIONS

8.1. Recursive utility

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$. The retained Epstein–Zin normalization is

$$f(c, V) = \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \quad (22)$$

the appropriate logarithmic limit when $a = 0$; see Epstein and Zin (1989), Duffie and Weil (1992). Evaluation is $V_t + \mathcal{L}^{c,p}V + f(c, V) = 0$, not an additive-reward equation with preference parameters renamed. The transform $V = \exp(W)/b$ enforces the sign domain. Compact positive margin for bV , bounds on consumption, and a monotone implicit step are additionally needed for a finite Lipschitz bound on the evaluation operator.

The gain argument extends beyond affine evaluation. If the action operators $\mathcal{T}_n^a$ are monotone uniformly Lipschitz with constants L_n on a verified continuation-value interval, then (8) is with δ^{j-n} replaced by $\prod_{m=n}^{j-1} L_m$, and the terminal factor by $\prod_{m=n}^{N-1} L_m$. Appendix D makes this statement. A sign-enforcing architecture alone does not verify the required compact interval or monotonicity.

The supplementary record retains the stationary homothetic anchors $p = .3$ and $c/x = .0455$ as declared parameters, the executed implicit recursive reference, and the R5 trained recursive reference. In the latter, raw losses around .0425 and .0502 against a .01 target were corrected using a reference. Those corrected results establish finite-model performance of the corrected results, but they do not establish small correction cost or a scaled neural recursive-portfolio solution. These experiments and their limitations are preserved rather than replaced by an additive-control alternative.

8.2. Sophisticated temporal selves

For a sophisticated acting self with present-bias parameter β and ordinary discount δ , the improvement and evaluation equations differ:

$$c_n(s) \in \arg \max_c \{u(c)\Delta + \beta \delta \mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (23)$$

$$V_n(s) = u(c_n(s))\Delta + \delta \mathbb{E}[V_{n+1}(F(s, c_n(s), \varepsilon))]. \quad (24)$$

Improvement carries the extra β . With two consumption dates, terminal log wealth, zero interest, and $\delta = 1$, the sophisticated first-date share is $1/(1 + 2\beta)$, hence .4166667 at $\beta = .7$. Replacing ordinary evaluation by geometric- β evaluation gives .4566210, a different behavioral model. The retained derivation, executed residual check, and feasible-deviation calculation are in part of the supplement. Equilibrium verification concerns each acting self's feasible evaluation with future selves fixed, not a fictitious commitment planner who can change all future selves' preferences.

player i , fix the rivals' feedbacks and solve both the induced policy-evaluation problem and the unilateral best-response problem over all subsequent dates and states. Applying the finite continuous gain certificate to this best-response problem bounds player i 's exploitability. A certificate requires this test for every player. Differentiating joint profits, allowing gradients to flow through rivals, or checking only realized-path deviations is a different operation. The retained static Cournot and three-date stochastic capacity-game records test these distinctions. The dynamic game has two firms, capacity investment, stochastic demand, and multiple equilibrium selections. The R5 neural game adds actual actor training, but its eighteen states and analytic best-response verification remain small-scale. Neither experiment establishes general equilibrium scalability or a competitive limit. The operator formulation, nonzero present value, recursive aggregator, and strategic state transitions are retained as substantive extensions; certified experiments on these objects remain a separate quantitative requirement, not a consequence of the new additive nonconvex benchmark.

9. COMPUTATION WITH STOCHASTIC TRACES

If A is symmetric and z_k are independent standard Gaussian probes, put $\bar{q}_K = K^{-1} \sum_k z'_k A z_k$.

$$\mathbb{E}(c - \bar{q}_K/2)^2 = (c - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (25)$$

Using separately squared one-probe residuals retains a variance penalty that does not decrease with K . Two independent probe means give an unbiased cross-product estimator of the squared exact residual, but not a nonnegative sample objective or a uniform residual bound. The proof, variance accounting, and the retained Monte Carlo regression are given in the Supplement and Appendix E.

For diffusion loading with m columns, an exact contraction can use m directional second-order tests; a randomized contraction uses K probe operations and the loading operations. Rank size, parameter differentiation, collocation coverage, optimization iterations, and the m -dependent probe budget all enter total cost. Avoiding a dense Hessian is useful, but it does not imply a universal dimensional crossover or eliminate the cost of verification. The coupled linear-quadratic tests with off-diagonal drift, rewards, and shocks are useful for model checks, while their known quadratic value family and faster Riccati comparator support a general performance conclusion.

10. CONCLUSION

The framework makes a policy-evaluation approximation, an improvement rule, and a verification method on separately inspectable. In a stopped preference-portfolio economy, this separation supports a complete constrained Hamiltonian audit, one-sided continuous value witnesses, and verifiable conditions for adjustment-cost comparisons at finite numerical accuracy. The new geometry certificate and paired author-code comparison address different questions: the former of a verification method and realized performance on coupled nonconvex controls. The latter is used to erase the conservative original-payoff certificate or the substantial cost of a safeguard.

The economic scope continues to include recursive utility, sophisticated temporal selves, and strategic interaction, with their distinct evaluation and equilibrium conditions. The remaining computational task is concrete: construct sufficiently narrow original-model witnesses, or enclose

APPENDIX A: PROOFS OF THE CONTINUOUS VERIFICATION RESULTS

A.1. Proof of Theorem 2

or a smooth witness w and a fixed admissible policy a , localized discounted Itô calculus
s

$$\left[e^{-\rho(\sigma-t)} w(\sigma, S_\sigma) + \int_t^\sigma e^{-\rho(r-t)} \ell(S_r, a_r) dr \right] = w(t, s) + \mathbb{E} \int_t^\sigma e^{-\rho(r-t)} R_{a_r} w(r, S_r) dr, \quad (26)$$

or σ first stops on a compact interior localization or at the economic stopping time. Localiza-
also bounds the stochastic integral. The hypotheses on the witness and uniform integrability
nit passage to τ ; no regularity of the optimal value is invoked.

or $w = U$, $R_a U \leq \epsilon_U$ for every admissible action. The terminal inequality $g \leq U + b_U$ and
 ≤ 1 yield

$$J^a(t, s) \leq U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

the supremum over policies; attainment is unnecessary. For $w = L$, use the specified policy
the inequalities $R_\pi L \geq -\epsilon_L$ and $g \geq L - b_L$, and the same integration identity to obtain the
er bound. Subtraction proves the regret bound, and nonnegativity follows from the definition
the supremum.

or multiple time slabs, apply (26) on each slab and sum. Forward-time jumps contribute
 $U(t^+, S_t) - U(t^-, S_t)$ to the upper-witness identity. They are nonpositive under Assumption 1,
the upper inequality is preserved. Lower-witness jumps have the opposite sign. This explains
a piecewise-time approximation may not silently omit its interface errors. The actual policy
be only piecewise continuous in time because Itô's formula is applied to the witnesses, not
derivative of the policy. *Q.E.D.*

A.2. Spatial residual potentials

suppose the upper residual satisfies $R_a U \leq q_U$ and a nonnegative potential obeys

$$z_{U,t} + \mathcal{L}^a z_U - \rho z_U \leq -q_U \quad \text{for every } a.$$

on $R_a(U + z_U) \leq 0$ by the affine dependence of R_a on the witness. Its boundary inequality is
erved by nonnegativity of z_U . Similarly, if $R_\pi L \geq -q_L$ and $z_{L,t} + \mathcal{L}^\pi z_L - \rho z_L \leq -q_L$, then
 $L - z_L \geq 0$. Theorem 2 applies to $U + z_U$ and $L - z_L$. A potential solving an occupation
blem exactly would weight residuals by reachable discounted occupation; an approximate
ntial is usable only after the displayed differential inequality is enclosed. This construction
not turn an empirical occupancy histogram into a theorem.

A.3. Verification of the explicit original-economy witnesses

nce $\partial_t C_\rho(1-t) = -e^{-\rho(1-t)}$ and $\rho C_\rho(1-t) = 1 - e^{-\rho(1-t)}$,

$$(\partial_t - \rho)[-a C_\rho(1-t)] = a$$

On the stopped boundary,

$$-g = 8(1-t) - \alpha C_\rho(1-t) \geq (8-\alpha)(1-t) \geq 0, \quad L-g = -\beta C_\rho(1-t) \leq 0.$$

NBO: R10 PRESERVATION SUPPLEMENT

43

witnesses equal G at maturity. They are smooth and bounded on the closed state rectangle $x \geq .5$. The bilinear policy is globally Lipschitz on this compact rectangle, component-wise, and piecewise constant in its time index. The stopped coefficients are bounded and required state regularity, so the discounted stochastic integrals are integrable. The sign therefore proves a statement about the original continuously stopped policy, including for which the optimal boundary regularity conditions of a different PDE theorem fail. Interval calculation bounds all a in the upper witness through the exact maximizing G . It bounds the lower witness on every time-state cell by the convex range of the policy corners, enlarged to a Cartesian box; enlarging the action range can only make the bound more conservative. Interval dependency can likewise enlarge the bound, never narrower true range. No assertion of floating-point simulation conformance is needed: mathematical policy is defined by the stored constants and exact interpolation weights. A deployment-error allowance would be required for another implementation.

A.4. Proof of Proposition 1

action-dependent part of $R_a v$ is the sum of

$$\frac{c^{1-u}}{1-u} - v_x c, \quad v_u \theta - \frac{k}{2} \theta^2, \quad h_1 p + \frac{h_2}{2} p^2.$$

fixed state derivative changes h_1 but introduces no product between control components. Assumption objective has derivative $c^{-u} - v_x$ and strictly negative second derivative $^{-1}$. If $v_x \leq 0$, its derivative is positive everywhere. Otherwise its unique unconstrained vertex point is $v_x^{-1/u}$, and projection gives the constrained optimum. The adjustment objective is strictly concave quadratic because $k > 0$. The portfolio objective is strictly concave < 0 , linear when $h_2 = 0$, and convex when $h_2 > 0$; the stated vertex or endpoint rule is true in all cases. Compact product feasibility and additivity make these separate maximizers a joint maximizer. Q.E.D.

A.5. A learned-geometry regret bound

(8), let $H_g^*(t, s) = \sup_a R_a g(t, s)$ for the original running payoff. The modified residual $= R_a g - H_g^* \leq 0$, with equality for the feasible explicit optimizer. Applying (26) first to g and then to this optimizer proves $\tilde{V} = g$. For the learned policy,

$$g(t, s) - \tilde{J}^\pi(t, s) = \mathbb{E}_{t,s}^\pi \int_t^\tau e^{-\rho(r-t)} \{H_g^* - R_\pi g\}(r, S_r) dr.$$

action-independent terms cancel. At $k = 2$ the adjustment term completes the square to $(u-2)^2$. The other two action differences are exactly those bounded in Section 6.3. Interval-state interval upper bound times $C_\rho(1-t)$ gives the recorded continuous regret rate. Policy sup-norm errors are separate interval consequences, not inferred by reversing the inequality. Q.E.D.

B.1. The exact separated-operator limit

Let $V_j = \mathcal{E}\pi_j$ and $\pi_{j+1} = \mathcal{I}(V_j)$. The selector inequality implies $R_{\pi_{j+1}}V_j \geq 0$, so exact comparison and verification yield $V_{j+1} \geq V_j$. Precompactness of the evaluation image bounds the sequence. Its pointwise increasing limit is denoted $\bar{V}$.

Every subsequence has a further subsequence converging in the declared norm of the function, its derivative, gradient, and Hessian. The function component of any such limit must equal $\bar{V}$, by pointwise monotonicity. Its derivative components are the derivatives of that same function limit. Thus the whole value sequence converges in that strong norm: otherwise a subsequence extracted from the unique possible strong cluster point would contradict precompactness.

Along a policy subsequence $\pi_{j_m} \rightarrow p$, continuity of evaluation gives $\mathcal{E}p = \bar{V}$. Continuity of the selector at this cluster point gives $\pi_{j_m+1} \rightarrow \mathcal{I}(\bar{V})$, and continuity of evaluation gives $\mathcal{E}(\mathcal{I}(\bar{V})) = \bar{V}$. The evaluation equation of the improved limiting policy therefore reads

$$R_{\mathcal{I}(\bar{V})}\bar{V} = 0.$$

The defining maximizing property of that selector implies $\sup_a R_a \bar{V} = 0$. Boundary comparison identifies $\bar{V} = V^*$. The proof establishes equality of limiting values, not equality of consecutive limiting policies. Compactness, selector closure, and strong evaluation continuity are hypotheses, not properties of an arbitrary neural architecture or Adam trajectory. *Q.E.D.*

B.2. Proof of Theorem 1

Positivity and substochasticity give $\|\mathcal{T}_n^a v - \mathcal{T}_n^a w\|_\infty \leq \delta \|v - w\|_\infty$. The maximized operator has the same Lipschitz constant. Since the candidate set includes the enacted action,

$$\|v_n - \max_a \mathcal{T}_n^a v_{n+1}\|_\infty \leq e_n + \eta_n.$$

Comparing this approximate equation with optimal backward induction yields

$$\|v_n - V_{h,n}^*\|_\infty \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

Comparing instead with policy evaluation gives the same inequality with only e_j in the sum. Combining the two bounds to prove (8).

For the localized bound let $d_n = V_{h,n}^* - V_{h,n}^\pi$. Exact evaluation implies $g_n \geq 0$. For every n , the elementary maximum-difference inequality gives

$$d_n \leq g_n + \delta \max_a P_n^a d_{n+1}.$$

Starting with $d_N = 0$, induction and positivity give $d_n \leq B_n$. The reference optimal value is necessary in this recursion: only policy evaluation, transition matrices, and feasible gains are needed.

Q.E.D.

B.3. Threshold correction and arithmetic scope

Every recomputed gain is at most γ , Theorem 1 gives a regret bound at most $\gamma \sum_n \delta^n$. Thus the bound is sufficient with margin when this uniform condition holds. Correcting a subset of decisions does not by itself guarantee that condition, since the re-evaluated continuation changes. The

of replacements. Its stopping certificate is the actual envelope of the corrected policy. Proposition 2, let I_m indicate $g_m > \gamma$. Then $\gamma I_m \leq g_m$, so $M^{-1} \sum_m I_m \leq \sum_m g_m$. The fixed- and shrinking-threshold implications follow directly. This not measures the correction share under the audit preservation decomposition imply that state losses have small occupation-weighted averages or that a full audit is operationally free. *Q.E.D.*

finite envelopes reported in the numerical tables are float64 quantities with independent agreement and regression tests. They are not outward-rounded enclosures of every arithmetic operation. To make a machine-rigorous finite certificate one must add roundoff error-solve enclosures to e_n and η_n , or evaluate their analogues with validated arithmetic. Continuous witness and learned-geometry certificates do use outward interval arithmetic, identified separately for this reason.

B.4. Proof of the continuous semigroup error decomposition

settlement fixed inside each step, the exact surviving-continuation map has Lipschitz at most δ : survival probabilities are at most one and continuation is received at the end. Taking a supremum over all admissible within-step controls preserves this bound. $E_n^* = \|V_n - w_n^*\|_\infty$ and $E_n^\pi = \|J_n^\pi - w_n^\pi\|_\infty$. Then

$$E_n^* \leq \kappa_n^* + \delta E_{n+1}^*, \quad E_n^\pi \leq \kappa_n^\pi + \delta E_{n+1}^\pi.$$

ard iteration and the positive norm-one lift of the finite gap prove (15). If $\mathcal{B}^{(0)}, \dots, \mathcal{B}^{(m)}$ intermediate operator chain from the exact semigroup to the lifted numerical operator, single inequality bounds its discrepancy on a specified continuation w by $\sum_i \|(\mathcal{B}^{(i)} - v)\|_\infty$. Every term must have a verified bound; changing the tested continuation between differences without controlling that change invalidates the telescoping argument. *Q.E.D.*

APPENDIX C: PROOFS FOR ECONOMIC COMPARISONS

fixed policy, $A(\pi) - kB(\pi)$ is affine and nonincreasing in k . The supremum is therefore and nonincreasing. The budget is bounded by $.02C_\rho(T - t)$ because $|\theta| \leq .2$. If π attains at k , for every other price k' ,

$$V(k') \geq A(\pi) - k'B(\pi) = V(k) - (k' - k)B(\pi).$$

$B(\pi)$ is a subgradient of V at k . At a differentiability point every attained optimizer has $-V'(k)$.

C.1. Proof of Theorem 3

proximate optimality and feasibility of the rival candidate policy give

$$A(\pi_1) - k_1 B(\pi_1) \geq A(\pi_2) - k_1 B(\pi_2) - \varepsilon_1,$$

$$A(\pi_2) - k_2 B(\pi_2) \geq A(\pi_1) - k_2 B(\pi_1) - \varepsilon_2.$$

and collecting terms yields $(k_2 - k_1)\{B(\pi_2) - B(\pi_1)\} \leq \varepsilon_1 + \varepsilon_2$. The exact monotonicity is the zero-error case. Adding the two budget measurement discrepancies yields and observable-error inequality. Feasible-set independence of k is essential: otherwise a used in the crossed comparison might not be available.

$= (\varepsilon_1 + \varepsilon_2)/(k_2 - k_1)$, $A_1 = 0$, and $A_2 = k_1 B_2 + \varepsilon_1$. At k_1 , policy 1 is exactly ε_1 suboptimal; policy 2 is exactly ε_2 suboptimal. Equality holds in the budget-reversal bound. Choose $\varepsilon_1, \varepsilon_2$ small enough to respect any prescribed positive upper bound on B_2 . This construction shows that the coefficient cannot be improved without more structure; it is not a claim that these artificial policies have been realized in the particular NDU dynamics. *Q.E.D.*

C.2. Proof of Proposition 3

Use the subgradient inequality with the optimizer at k and prices k_+ and k_- . Rearrangement gives

$$\frac{V(k) - V(k_+)}{k_+ - k} \leq B(\pi_k) \leq \frac{V(k_-) - V(k)}{k - k_-}.$$

Replacing each numerator by its appropriate outer value enclosure gives (17), and intersecting with the primitive budget range is valid. Inclusion of restricted policies in the flexible set gives $V_F - V_R \geq 0$. The interval subtraction rule gives the other endpoints. No optimizer differentiability, common selected policy, or pointwise action ordering is required. *Q.E.D.*

APPENDIX D: RECURSIVE AND EQUILIBRIUM GAIN BOUNDS

Suppose a verified interval of continuation values is invariant under all action operators, each operator is monotone on it, and each has Lipschitz constant at most $L_n \geq 0$. The maximized operator is also L_n -Lipschitz. Repeating the two approximate-equation comparisons in the finite horizon gives

$$\begin{aligned} \|V_n^* - v_n\|_\infty &\leq e_n + \eta_n + L_n \|V_{n+1}^* - v_{n+1}\|_\infty, \\ \|V_n^\pi - v_n\|_\infty &\leq e_n + L_n \|V_{n+1}^\pi - v_{n+1}\|_\infty. \end{aligned}$$

This proves the product-weighted error formula stated in the text. These hypotheses must be verified for the actual recursive discretization. In particular, the sign condition $bV > 0$ does not rule out arbitrarily large derivatives of the driver close to zero and does not supply a uniform L_n . For a game, fix rivals' strategies and regard player i 's unilateral deviations over all future dates as a control problem with its own payoff and transition law. The preceding certificate bounds $V_i^{\text{BR}(\pi_{-i})} - V_i^\pi$. Requiring this bound for every player gives an approximate Nash condition at every initial state included in the norm. Checking only a single-date action deviation on only states visited in one rollout does not replace the all-future-date best-response calculation. For sophisticated temporal selves the acting self's criterion uses the β -weighted continuation value (24), but ordinary continuation evaluation uses δ . A unilateral deviation certificate must use the acting self's criterion, with future selves held fixed. Substituting a geometrically β -discounted evaluation equation changes that equilibrium problem, so its small residual would certify the wrong object. The preserved supplement supplies the full two-date algebra and the unrolled finite-model conventions. *Q.E.D.*

APPENDIX E: TRACE ESTIMATION AND LIMITATIONS OF LOCAL TRAINING

For a symmetric matrix A and $z \sim N(0, I)$, diagonalize A orthogonally. Rotational invariance and independence of the coordinates imply $\mathbb{E} z' A z = \text{tr } A$ and $\text{Var}(z' A z) = 2 \|A\|_F^2$. The mean

25).
 ing the K quantities $(c - z'_k A z_k / 2)^2$ leaves the expectation equal to $(c - \text{tr } A / 2)^2 +$
 . With two independent probe means $\bar{q}$ and $\bar{q}'$, independence instead gives

$$\mathbb{E}[(c - \bar{q}/2)(c - \bar{q}'/2)] \stackrel{\text{NBO: R10 PRESERVATION SUPPLEMENT}}{=} (c - \text{tr } A / 2)^2.$$

ter sample objective may be negative and may have high variance. None of these
 ion identities is a uniform residual enclosure over state space. *Q.E.D.*
 al actor stationary point also cannot substitute for a global feasible action maximum.
 $u) = -(a^2 - 1)^2 + .2a$ on $[-2, 2]$, the local maximum near $-.973994$ and the global
 m near 1.024120 are both isolated stationary points, with objective gap about $.399875$.
 ritic evaluation and isolated actor stationary points do not eliminate this difference.
 ty in the action would give the global supporting-hyperplane gap, but concavity in an
 rgument does not imply concavity in shared network weights.
 etained viscosity counterexample has the same logical role. A small integrated strong
 for a smooth approximation of $|x| - 1$ does not select the viscosity solution $1 - |x|$ of
 opriate eikonal boundary problem. Missing viscosity inequalities cannot be replaced by
 ot. The previous derivations, exact boundary specifications, and regression checks for
 amples are reproduced unchanged in the preservation supplement.
 ise, a two-timescale stochastic-approximation theorem requires bounded iterates, ap-
 e regularity, stable fast evaluation dynamics, martingale-noise control, and step-size
 ns. Its generic conclusion is an invariant-set statement for the limiting slow dynamics.
 nal-policy conclusion needs an additional argument ruling out nonoptimal attractors.
 rent regret theorems do not rely on asserting that every observed gradient trajectory has
 optimal limit.

47

Dynamic economic models increasingly place the formation of preferences, the allocation of financial risk, and the adjustment of physical resources inside the same decision problem. The relevant computational object is then a controlled operator: a policy changes both the payoff and law under which continuation utility is evaluated. A useful approximation must preserve this distinction. In particular, reducing a joint scalar training objective need not improve an economic policy, and a small error in a value representation need not identify the policy at a globally flat optimum.

This paper develops Neural Bellman Operators (NBO) as a separated evaluation–improvement method for such problems. The critic approximates the continuation value of a specified feasible policy. The actor improves that policy against a frozen continuation object. A third, independent evaluation evaluates the enacted policy and its feasible deviation gains. We study both the continuous generator formulation and a positive finite-horizon operator formulation. The latter treats the stochastic transition and the action maximization explicitly and provides a reference against which a trained neural representation can be assessed.

There are three substantive results. First, we give a quantitative connection between policy evaluation, boundary approximation, feasible improvement, and loss of economic value. The continuous-time bound applies to a stopped controlled diffusion and requires uniform, rather than merely sampled, residual bounds. Its finite-horizon counterpart is directly computable as a finite state–action approximation. Second, we establish a comparative-static result for endogenous preference formation. When the adjustment cost enters as a price of a discounted settlement budget, an increase in that price reduces the optimal budget. This conclusion holds for any optimizers and does not require differentiability of the value function or a pointwise restriction on preference adjustment. Third, we implement the separated method, including automatic differentiation of a continuous HJB benchmark and trained nonlinear actor and critic networks for a stochastic preference–portfolio problem. Independent policy evaluation separates the accuracy from agreement of the full policy vector.

The economic application preserves the interior preference and wealth dynamics of the final model. It completes the model at the boundary by an explicit early-liquidation contract. The preference coefficient has nonzero diffusion, so a bounded drift cannot keep it in a closed interval with probability one. Likewise, the original consumption floor is incompatible with a financing wealth process constrained to the lower endpoint. Liquidation is an economically relevant boundary condition from reflection or a capital injection; it is specified as such, including the settlement fee and its sensitivity. No transition replenishes wealth and then resumes consumption.

The numerical evidence has distinct roles. A continuous-time Merton benchmark tests the HJB derivatives and the separated gradient graph. A two-state NDU problem tests a trained nonlinear neural representation against an independent exhaustive stochastic Bellman solver. A re-solved settlement-cost panel evaluates the economic comparative static. Recursive utility, sophisticated moral selves, and a dynamic capacity game exercise additional operator interfaces. A coupled stochastic linear–quadratic problem checks dynamic cross-state dependence at dimensions 4, 8, and 16. These experiments retain the general economic scope without conflating an algebraic identity, a finite-model equilibrium, and a trained solution of a continuous nonlinear economy.

1.1. *Relation to the literature*

The closest methodological antecedents already combine policy improvement and neural approximation. Kim, Kim, Kim, and Cho (2025) study physics-informed neural policy iteration

erator learning. Cai, Fang, and Zhou (2025, preprint) learn value and control through a game formulation without requiring an explicit Hamiltonian optimizer. Direct neural approximation of stochastic control policies is developed by Han and E (2016). Continuous-time improvement and its hypotheses are analyzed by Jackarand Mijatović (2015). Accordingly, approximation, operator factorization, and the absence of an analytical maximizer are not here as inventions in themselves.

Contribution relative to these approaches is the particular combination of boundary-aware asset control, an economically interpretable preference-adjustment result, and an executable solution of the evaluation, improvement, and policy-certification objects. The finite-operator approach also permits a distinction between errors of neural representation and errors of the real economy. The bounds below are not presented as a priority claim for residual-based solution in general; they specify the conditions and constants used by this implementation application.

Economic formulation connects portfolio choice (Merton, 1969), recursive preferences (Merton and Zin, 1989, Duffie and Epstein, 1992), and the motivating endogenous-preference framework of Qi (2025). Grid and projection methods remain important comparators in computational finance (Judd, 1998, Brumm and Scheidegger, 2017). In our two-state experiment, exhaustive grid induction is faster than neural training. That result is reported, not converted into a claimed scalability advantage. A neural representation can nevertheless be evaluated as an approximation to a controlled operator, and an independently measured value loss can be a meaningful test of that representation.

A distinction from Deep BSDE methods (Han, Jentzen, and E, 2018) is about the equations defining objects, not about whether an analyst can write a closed-form control. Semilinearity is dependence on the highest-order derivative. Eliminating a control analytically can remove nonlinear Hessian dependence. Conversely, an admissible BSDE representation need not solve the adjoint system. Section 7 states the differentiation costs conditionally, including dimension rank and the accuracy demanded of a trace approximation.

2. THE ECONOMIC PROBLEM AND ITS BOUNDARY

Let s be a Markov state in an open bounded domain $D \subset \mathbb{R}^d$, let $A(s)$ be a compact feasible correspondence, and let π be an admissible feedback. Up to $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$, S satisfies

$$dS_r = b(r, S_r, \pi_r) dr + \Sigma(r, S_r, \pi_r) dW_r. \quad (1)$$

Existence of the stopped process and integrability of the stated payoff. For a separable payoff with nonnegative discount rate ρ , define

$$J^\pi(t, s) = \mathbb{E}_{t,s} \left[\int_t^\tau e^{-\rho(r-t)} \ell(r, S_r, \pi_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V^* = \sup_\pi J^\pi. \quad (2)$$

The element g is data of the problem, not an output of a projection routine. At T , it equals the terminal utility. The operator is

$$\mathcal{L}^a v = b^a \cdot \nabla v + \frac{1}{2} \text{tr}(\Sigma^a (\Sigma^a)' D^2 v), \quad R_a(v) = v_t + \mathcal{L}^a v + \ell^a - \rho v.$$

If a classical representation is valid, fixed-policy evaluation has $R_\pi(V^\pi) = 0$, while optimality has $\sup_a R_a(V^*) = 0$. Both have the same declared parabolic-boundary data. A reflected BSDE would instead include regulators and an appropriate derivative boundary condition; the BSDEs are not obtained by replacing that condition with a value penalty.

ended state process; the payoff and stochastic integrals used below are integrable. For each policy under discussion, evaluation is well defined. Comparison and the relevant verification inequalities hold for the specified generator and boundary operator. Whenever derivatives are used, the test function belongs to $C^{1,2}$ up to the relevant stopped boundary and its discounted formula is valid.

These hypotheses state the mathematical object to which a residual refers. They do not assert an arbitrary smooth network or an arbitrary compact computational rectangle verifies them.

2.1. Recursive utility and Markov closure

For recursive preferences, fixed-policy utility is defined before optimization. On a domain of value argument where the driver is admissible, it solves

$$-dY_r^\pi = f(r, S_r, \pi_r, Y_r^\pi) dr - Z_r^\pi dW_r, \quad Y_\tau^\pi = g(\tau, S_\tau). \quad (3)$$

Existence, uniqueness, comparison, and integrability of this policy-wise equation are prerequisites for taking its supremum. The Markov state must contain any variable on which the present or future opportunities depend. A general continuous martingale is not replaced by a Gaussian shock merely because its quadratic variation is known; the executed models here use stated Brownian or finite Markov specifications. The recursive extension and its sign domain developed in Section 6.1.

3. SEPARATED NEURAL BELLMAN OPERATORS

In the continuous realization, a critic v_ξ is trained for a fixed enacted policy by its evaluation residual and the appropriate boundary loss. The actor parameters ψ are updated against a trained critic jet:

$$L_E(\xi; \pi) = \mathbb{E}_\nu R_\pi(v_\xi)^2 + \lambda_B \mathbb{E}_{\nu_B} |v_\xi - g|^2, \quad (4)$$

$$L_I(\psi; \xi) = -\mathbb{E}_{\nu_I} \mathcal{H}^{\pi_\psi} [\text{stopgrad}(v_\xi, \nabla v_\xi, D^2 v_\xi)]. \quad (5)$$

Feasibility is imposed by the actor parameterization or by explicit constrained maximization. A penalty for a general constraint is an approximation and must be accompanied by a feasibility check. During actor improvement no parameter gradient reaches ξ . In a game no player- i gradient reaches the parameters of a frozen rival.

The actor loss is a local improvement surrogate. It is not generally the lifetime-objective gradient under an arbitrary sampling measure. Such an identification requires the appropriate occupancy or adjoint weighting and differentiability conditions. Likewise, minimizing a weighted sum of (4) and (5) over both parameter blocks solves a different optimization problem. Appendix E retains the composite-loss and viscosity counterexamples and states what the approximation argument does establish.

3.1. A positive finite-operator realization

For the numerical economy, let $\mathcal{S}_h$ be a finite state set and A_h a finite feasible action set. Identification proceeds are included in r_n^a and surviving transitions in a substochastic kernel P_n^a . A discount $\delta = e^{-\rho\Delta}$,

$$T_n^a w = r_n^a + \delta P_n^a w, \quad T_n w = \max_{a \in A_h} T_n^a w. \quad (6)$$

in r_n^a . Boundary states are settled immediately.

finite NBO algorithm works backward. It freezes the next critic, computes each feasible continuation using the declared stochastic operator, and trains the current actor against action values. It then trains the current critic to evaluate the actor's sampled hard action. A numerical experiment uses a cross-entropy term for the maximizing action of the *neural operator*, together with its softmax-weighted action-value regret. Optimal reference values never enter this training loss. At deployment the actor chooses its hard argmax, and the critic computes the softmax-weighted action-value regret. All continuation targets are detached. This is a finite-horizon neural Bellman operator, not a claim that one Adam update performs exact policy iteration.

A reference solver is an independently coded NumPy backward induction with exhaustive evaluation and positive bilinear interpolation. The neural operator is implemented separately in PyTorch. An operator-agreement test precedes training. The trained policy is then evaluated by the reference operator; this evaluation is different from reading the critic's output. All maxima used for finite-model certificates range over the entire stated finite set.

3.2. Exact improvement and quantitative approximation

PROPOSITION 1—Exact separated-operator limit: *Let the policy space Π be compact in the topology. Let evaluation E be continuous into a precompact subset of $C^{1,2}$, endowed with the uniform norm on the function, its time derivative, gradient, and Hessian. Assume a selector I maps Π into itself, is continuous at accumulation points, and satisfies the consistency and verification conditions of Assumption 1. If $\pi_{j+1} = I(\pi_j)$, every accumulation point p satisfies $E(p) = V^*$, and the common limiting value solves the HJB boundary problem.*

The decisive step is equality of limiting values, not equality of successive limiting policies. The proof is in Appendix A. Closure under the exact selector is a real hypothesis and is not automatic for a finite neural class.

THEOREM 1—Boundary-aware policy-value certificate: *Under Assumption 1, let v be a test function and π a feasible policy. Suppose, uniformly on the stopped domain,*

$$|v| \leq \epsilon, \quad 0 \leq \sup_{a \in A(s)} \{R_a(v) - R_\pi(v)\} \leq \eta, \quad |v - g| \leq b \quad \text{on the parabolic boundary.}$$

with $C_\rho(h) = (1 - e^{-\rho h})/\rho$ and $C_0(h) = h$,

$$0 \leq V^*(t, s) - J^\pi(t, s) \leq 2b + C_\rho(T - t)(2\epsilon + \eta). \quad (7)$$

Thickness of V^ , and no claim that an actor stationary point is a maximizer, is required for equality.*

The boundary term is not discounted at the terminal date: exit can occur earlier. Theorem 1 decomposes separately meaningful errors into a value loss. A sampled mean-square loss does not satisfy the uniform hypotheses. A verified residual Lipschitz constant and a covering radius can be used to convert a finite audit to a uniform bound; such constants must be established, not inferred from a

THEOREM 2—Finite-horizon neural-policy certificate: *For the finite operators (6), let $\hat{v}_N$ be the terminal error at most b . For an enacted policy π , define*

$$e_n = \|\hat{v}_n - T_n^\pi \hat{v}_{n+1}\|_\infty, \quad \eta_n = \|T_n \hat{v}_{n+1} - T_n^\pi \hat{v}_{n+1}\|_\infty.$$

$$\|V_n^* - V_n^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

52

the enacted policy is re-evaluated exactly on this finite model, take $\hat{v} = V^\pi$; then $e_j = b = 0$ the same bound uses only the independently evaluated feasible improvement gaps.

This theorem bounds a finite model, not automatically the underlying diffusion. Time, interpolation, quadrature, action coverage, and boundary-crossing errors remain distinct. An additional needed comparison between the finite and continuous economies would be needed to add them up).

3.3. Global action improvement versus parameter stationarity

The Hamiltonian $H(a) = -(a^2 - 1)^2 + .2a$ on $[-2, 2]$ has a stable local ascent attractor near .73994 and a global maximum near 1.024120. Their payoff gap is .3998746. Both stationary points are isolated. Thus an exact fast critic, diminishing two-timescale steps, and isolated actor stationary points do not imply a policy-improvement fixed point.

A usable sufficient condition concerns the *action* maximization. If H is differentiable and concave on a convex compact feasible set, then

$$H^* - H(a_0) \leq \max_{a \in A} \langle \nabla H(a_0), a - a_0 \rangle. \quad (9)$$

The right-hand side is a global supporting-hyperplane gap and handles boundary optima. Point-concavity does not imply concavity in shared neural parameters. Our finite implementation therefore audits the maximizing finite action directly. The separate asymptotic stochastic-approximation statement, with its invariant-set conclusion and additional optimality conditions, given in Appendix E.

4. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

The preference state u is the coefficient of relative risk aversion and X is financial wealth. Controls are consumption c , preference adjustment θ , and the risky portfolio share π . Before randomization,

$$du = \theta dt + .05 dZ^u, \quad (10)$$

$$dX = [(.02 + .06\pi)X - c]dt + .2\pi X dZ^X, \quad d\langle Z^u, Z^X \rangle = -.25 dt. \quad (11)$$

The domain is $(1.2, 2.8) \times (.5, 2)$ and the control box is $[.05, .8] \times [-.2, .2] \times [-.5, .8]$. The horizon is one year and $\rho = .04$. Utility is cardinally fixed as

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad G(u, X) = -.02(u-2)^2 + .1 \log X.$$

Assumption units and this normalization are held fixed throughout the cost panel. An affine transformation that depends on the controlled preference state generally changes the economic problem.

process stops when it first reaches the domain boundary or the horizon. Settlement is

$$g(t, u, X) = G(u, X) - F(1 - t), \quad F = 8. \quad (12)$$

NBO: R10 PRESERVATION SUPPLEMENT

53

termination fee represents the utility cost of closing the investment and preference-arrangement before maturity. This is an explicit covenant/liquidation specification, calibrated estimate of such a fee. Its size is varied separately below. The fee vanishes at $t = 1$ and does not depend on k . No additional wealth is supplied after settlement because the investment has terminated.

completes rather than disguises the boundary economics. Constant preference diffusion at the earlier viable-rectangle interpretation. At $X = .5$, even the largest wealth drift in the original action box is $-.016$. Keeping the original interior process and the original floor therefore requires exit, reflection with an explicit regulator, or a different resource allocation. We choose exit. Economic statements about this contract must not be interpreted as statements about a silently reflected self-financing economy.

interior HJB is

$$0 = V_t + \sup_{c, \theta, \pi} \left\{ \frac{c^{1-u}}{1-u} - \frac{k}{2} \theta^2 + \theta V_u + [(.02 + .06\pi)X - c]V_X - \rho V \right. \\ \left. + \frac{1}{2}(.05)^2 V_{uu} + \frac{1}{2}(.2\pi X)^2 V_{XX} - .25(.05)(.2)\pi X V_{uX} \right\}, \quad (13)$$

on the parabolic boundary. All three second-order contributions are retained.

4.2. Policies and economic interpretation

the maximizer is interior and the indicated derivatives have the required signs,

$$c = V_X^{-1/u}, \quad \theta = V_u/k, \quad \pi = -\frac{.06V_X}{.2^2 X V_{XX}} - \frac{.05(-.25)V_{uX}}{.2X V_{XX}}. \quad (14)$$

are candidates, not substitutes for constrained maximization. Consumption and adjustment are concave action objectives. The portfolio objective is concave only when $V_{XX} \leq 0$; the endpoint comparisons are essential. Appendix B gives the full selection rule, including curvature and nonpositive marginal wealth value.

cross derivative produces a hedge against preference shocks correlated with returns. Its endogenous: neither a negative shock correlation nor an assumed positive V_{uX} alone makes a global portfolio comparative static. Likewise, dividing V_u by k does not determine the re-solved derivative changes with k . The following result gives a global economic conclusion without either shortcut.

LEMMA 3—The price of preference adjustment: *Suppose the feasible policy set, stopping and settlement contract do not depend on k . Write*

$$A(\pi) = \mathbb{E} \left[\int_t^\tau e^{-\rho(r-t)} \frac{c_r^{1-u_r}}{1-u_r} dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(r-t)} \frac{\theta_r^2}{2} dr.$$

$A(k) = \sup_\pi \{A(\pi) - kB(\pi)\}$ is convex and nonincreasing. If $k_1 < k_2$ and optimal π_1, π_2 exist, $B(\pi_2) \leq B(\pi_1)$. At a differentiability point with an attained optimizer, $A(k_1) - B(\pi_{k_1}) \geq A(k_2) - B(\pi_{k_2})$. The flexible-preference value weakly exceeds the value under the feasible policy when $\theta \equiv 0$.

5. NUMERICAL IMPLEMENTATION AND RESULTS

5.1. Continuous HJB benchmark

We first solve the stationary Merton economy with CRRA coefficient 2, discount .04, riskless
 return .02, excess return .06, and volatility .2. A homothetic critic is $v(x) = -\exp(w)/x$ and
 a logit actor represents $m = c/x \in [.005, .07]$ and $\pi \in [-.5, 1]$. First and second wealth
 derivatives are obtained by automatic differentiation. Critic optimization minimizes the scaled
 full-policy HJB residual. Actor optimization holds these derivatives fixed. Neither loss receives
 analytical optimal controls or value as a label.

Independent evaluation gives $A^\pi = 1/(mD)$ for $J^\pi = -A^\pi/x$, where $D = .04 + .02 +$
 $c - m - .04\pi^2 > 0$. The optimum has $m^* = .04125$, $\pi^* = .75$. Across seeds 0, 1, and 2
 the executed actor recovers m within 8.25×10^{-8} and π within 8.29×10^{-10} . The maximum
 independently evaluated value loss on $[.5, 2]$ is 4.70×10^{-9} . The scaled HJB residual is at most
 1×10^{-8} . A graph assertion verifies that actor backpropagation leaves the critic gradient
 connected. This is a homothetic neural benchmark with three trained parameters, not evidence
 of a deep high-dimensional approximation theorem. Its value is that the continuous generator,
 automatic derivatives, evaluation, and actor graph can all be checked independently.

5.2. Stochastic NDU transition and neural architecture

For the preference–portfolio model, four equiprobable branches use independent signs $z_1, z_2 \in$
 $\{-1, 1\}$:

$$\Delta u = \theta \Delta + .05\sqrt{\Delta}z_1, \quad (15)$$

$$\Delta X = [(.02 + .06\pi)X - c]\Delta + .2\pi X\sqrt{\Delta}[-.25z_1 + \sqrt{1 - .25^2}z_2]. \quad (16)$$

On a straight segment exits, it settles at its first intersection and terminates; otherwise con-
 tinuation uses positive bilinear interpolation. The running payoff is integrated with the exact
 discount factor for the elapsed fractional step while holding the current flow fixed. This defines
 a controlled-Euler numerical economy, not exact Brownian first-passage integration. Interior drift
 covariance tests are run before boundary corrections.

The nonlinear actor and critic each have two hidden layers of width 48 with tanh activations.
 The actor has 125 outputs, representing five values for each of the three controls. The critic uses
 normalized states and log distances to the four boundary faces; it is represented as $G + (1 - t)N_\xi$
 away from boundary nodes, with exact prescribed settlement at boundary nodes. The finite
 element does not require this boundary-switched representation to be a globally C^2 function.
 Consequently, the numerical certificate applied to it is Theorem 2, not an unverified use of
 Theorem 1.

At each of six dates, training uses all interior nodes of a 25×31 lattice, 500 actor Adam steps
 and 1,000 critic Adam steps, followed by critic L-BFGS refinement. The learning rate is .004.
 Past-date critics and actor targets are frozen. This experiment is a lattice-trained finite-operator
 approximation, not a mesh-free sampling experiment. Seeds 0, 1, and 2 are reported without selecting
 the best seed.

Quantity	Seed 0	Seed 1	Seed 2
Maximum date-zero value loss	.011291	.023526	.012797
Mean interior date-zero value loss	.001513	.001871	.001418
Maximum critic-policy value difference	.144195	.070268	.104457
Finite gain-based upper bound	.259331	.264491	.520503
Training seconds	19.039	19.609	20.737

compare the independently re-evaluated hard actor with exhaustive optimization on the identical finite economy. The gain bound uses maxima at all dates and is conservative. Times are local single-thread CPU measurements, not universal complexity constants.

Three completed runs meet the stated finite-model date-zero value-loss target of .05. The critic's own error is larger than the enacted policy loss, illustrating why a critic prediction is not a policy-evaluation certificate. Boundary discrepancies are at most 4.80×10^{-7} in the 32 neural output. For seed 0 the maximum component differences in (c, θ, π) from the optimal action are (.375, .2, .975); the corresponding interior mean differences are (.00335, .11028). The portfolio policy is therefore not uniformly accurate even when the regret is small. Every seed's complete component vector is retained in the ledger.

Exhaustive reference backward induction takes about .60 seconds in the seed-0 comparison, .9.04 seconds for training and .65 seconds for independent policy evaluation. There is no speed advantage in this two-state task. The result instead establishes that a genuinely trained actor and critic implement the stated stochastic operator with small independently calculated finite-model policy loss. Whether representation reuse or higher-dimensional nonlinear models reverse the cost comparison requires matched experiments on those objects.

The initial neural pilot failed: its maximum policy value loss was 4.10, its critic error 4.49, and its boundary representation was inaccurate. The failure and its executed source are retained. The next attempted twelve-date runs reached training output but timed out before completing their execution; they are recorded as incomplete, not successful runs. These records are not included in the three-seed success count.

5.3. Re-solved preference-adjustment costs

The cost panel uses twelve dates, the 25×31 state lattice, and the 125-action stochastic economy. Each k is independently re-solved, along with the restricted $\theta = 0$ economy. Table II shows the results for $(u, X) = (2, 1.25)$. The budget falls by approximately 86 percent between $k = .5$ and $k = 8$.

TABLE II

COSTLY PREFERENCE FORMATION UNDER THE LIQUIDATION CONTRACT

	$k = .5$	$k = 2$	$k = 8$
Value	-1.351522	-1.372708	-1.404420
Discounted adjustment budget B	.016520	.011921	.002309
Value of permitting adjustment	.064480	.043293	.011582
Date-zero adjustment θ	.2	.2	.1
Exit probability	.248935	.249938	.249920
Interior share with nonzero θ	.896927	.764993	.475637

The budget includes the factor $\theta^2/2$ and time discounting. Exit probabilities refer to this numerical liquidation contract. Nonzero-action shares average over all decision dates and interior lattice states, not over a population.

at all audited finite states and dates, up to the implementation tolerance. The preference gain is economically active rather than grid-locked. The result is conditional on the cardinal normalization and settlement contract; it is not identification of these primitives from observed consumption portfolios.

5.4. Discretization and boundary sensitivity

Separate refinements are deliberately not labeled solution-error certificates. At the same normal state, values for (N, n_u, n_X, n_a) are

$(6, 13, 16, 5)$	-1.423383
$(12, 13, 16, 5)$	-1.564151
$(12, 25, 31, 5)$	-1.372708
$(24, 25, 31, 5)$	-1.451486
$(24, 49, 61, 5)$	-1.344458
$(12, 25, 31, 7)$	-1.359628

where n_a is the number of values per control. The changes are material. Refining time alone while holding a positive interpolation grid fixed need not improve a diffusion approximation: the interpolation contribution to a generator can scale as h^2/Δ . Action refinement and boundary-crossing approximation also matter. We therefore report a verified finite-model neural approximation and a solved finite-model cost panel, but not a numerically certified continuous-diffusion value. The analytical comparison in Theorem 3 does apply to the continuous contract under its standard existence conditions.

Changing the settlement fee from 8 to 6 or 10, with the twelve-date baseline discretization held fixed, changes central value to -1.363909 or -1.379622 . These are different contracts rather than alternative numerical corrections to the same contract. The complete exit and adjustment calculations are retained so that subsequent boundary refinements can be evaluated without requiring post-correction feasibility as a zero-error result.

6. RECURSIVE UTILITY AND EQUILIBRIUM EXTENSIONS

6.1. Recursive utility

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$. In the normalization used here, the Epstein–Zin continuous-time aggregator is

$$f(c, V) = \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \quad (17)$$

the logarithmic limit when $a = 0$. The architecture $V = \exp(W)/b$ enforces the sign constraint when $b \neq 0$. A compact positive margin for bV , along with consumption bounds, implies local driver regularity; a sign assertion without executing the transform is not a domain constraint.

The full portfolio evaluation equation is $V_t + \mathcal{L}^{c, \pi} V + f(c, V) = 0$. Formal stationary homothetic ratios at $(\rho, \gamma, \psi, \mu, r, \sigma) = (.04, 5, 1.5, .08, .02, .2)$ are $\pi = .3$ and $m = .0455$. An algebraic solution is distinct from a verified infinite-horizon solution. We retain these anchors and add an audited finite-horizon implicit recursive-utility calculation. The latter has horizon .5, ten dates, wealth points and 80 consumption fractions, and solves each fixed-action implicit value equation on its negative-value branch. Its maximum implicit-equation residual is 2.46×10^{-13} ; minimum bV is .0625; the executed terminal sign transform differs from the terminal target

6.2. *Sophisticated temporal selves*

crete dates and $\delta = e^{-\rho\Delta}$, a sophisticated acting self takes the continuation policy of
selves as given:

$$c_n^*(s) \in \arg \max_c \{u(c)\Delta + \beta\delta\mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (18)$$

$$V_n(s) = u(c_n^*(s))\Delta + \delta\mathbb{E}[V_{n+1}(F(s, c_n^*(s), \varepsilon))]. \quad (19)$$

provement carries the extra factor β . For two consumption dates followed by terminal
lth, zero interest and $\delta = 1$, ordinary continuation has coefficient 2 on log wealth at
decision. The sophisticated initial consumption share is $1/(1 + 2\beta)$. At $\beta = .7$ this is
.7, whereas geometric- β evaluation gives .4566210. The corrected executed evaluation
is below 8.89×10^{-16} and a dense feasible one-shot search finds no positive gain. Strict
y and the analytical first-order condition establish global optimality for this particular
log example; the finite search alone would provide only a lower bound on exploitability.
1 check is retained but is not used to detect an error that vanishes at $\beta = 1$.

6.3. *Dynamic capacity competition*

ayer i , a frozen-rival policy defines both the evaluation operator and an independent
il best-response problem. The static Cournot benchmark has Nash output $1/3$ and joint-
output $1/4$; an automatic-differentiation test confirms that the rival gradient is absent from
s actor update.

so solve a genuinely dynamic finite game. Two firms have capacities in $\{0, .5, 1\}$, output
unit increments up to capacity, and investment $i \in \{0, 1\}$ that adds .5 capacity when
Demand intercept is $z \in \{2.5, 3.5\}$ with transition matrix having diagonal .8 and
onal .2. Per-period profit is $q_i(z - q_i - q_j) - .2i$, discount is .96, and the horizon is
roduction dates. Backward enumeration selects a pure Nash equilibrium in every state;
aphic selection resolves multiplicity in three state-date problems. The maximum feasible
t deviation gain is computed as zero in this finite game. The full profile and selection
stored, including asymmetric equilibria. This is not a neural dynamic-game training
a competitive-limit calculation. The latter would require a market-size normalization
eparately solved limiting economy; the operator formulation and its economic questions
ned in the appendix.

7. STOCHASTIC TRACES AND COUPLED DYNAMICS

ymmetric A and Gaussian probes z_k , write $q_k = z_k' A z_k$ and $\bar{q}_K = K^{-1} \sum q_k$. The
residual objective satisfies

$$\mathbb{E}(a - \bar{q}_K/2)^2 = (a - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (20)$$

g $(a - q_k/2)^2$ instead leaves a variance penalty independent of K . With $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ and
the desired expectation is $1 + 7.5/K$, not 8.5 for every K . The replication runs 30,000

red-objective mean checks fall within five reported standard errors. Appendix D gives an ased two-probe cross-product correction and explains why unbiased trace estimation does mply an unchanged training objective after squaring.

dense Hessian need not be materialized. For diffusion loading with m columns, the exact raction is a sum of m directional second derivatives; stochastic contraction requires K probe operations plus loading costs. Network size, parameter differentiation, sample count, imization iterations, and the probe budget needed for accuracy all enter total cost. There is model-independent crossover at state dimension 50 or 100.

nally, the coupled dynamic check has $d \in \{4, 8, 16\}$, eight dates, $x_{n+1} = Ax_n + a_n + S\varepsilon_{n+1}$, $.8I + .04\text{Adj}$, $Q = I + .1211'/d$, $R = .3I$, and $S = .1I + .0211'/d$. A linear neural actor mproved against a frozen quadratic policy evaluator, and compared with an independent ati solution. Off-diagonal transition, payoff, and shock terms are retained. Maximum -out value errors at the three dimensions are respectively 6.60×10^{-14} , 1.55×10^{-14} , and $\times 10^{-14}$. This is an unconstrained linear-quadratic dynamic regression, not the earlier static ratic program. Riccati evaluation is faster, and the known quadratic value family makes the structurally favorable. No constrained sparse-grid superiority or general high-dimensional plexity conclusion follows from it.

8. CONCLUSION

BO treats policy evaluation, feasible improvement, and independent economic evaluation as erent computational objects. The distinction yields explicit value-loss bounds and prevents itic residual or an actor stationary point from being mistaken for a solution certificate. he preference-portfolio application, an explicit liquidation contract makes the stochastic el and its numerical transition agree; the re-solved adjustment-cost panel implements bbal comparative static for the discounted preference-adjustment budget. Actual neural ates, policy evaluation, diffusion moments, temporal-self recursion, trace estimands, and -gradient conventions are now executable.

ne economic scope includes recursive utility and strategic interaction without assigning them tical mathematical hypotheses. Further accuracy must be earned through the declared error s: the current NDU runs certify the finite stochastic economy, while continuous-diffusion ement remains material. All historical models, derivations, figures, and prior reports remain e repository, with a retention map distinguishing corrected equations, supplementary erial, archival arrays, and executed evidence. This structure permits the next review to assess mathematics and the numerical economy rather than reconstruct their intended relationship.

APPENDIX A: PROOFS AND THE SCOPE OF THE VALUE CERTIFICATES

A.1. *Proof of Proposition 1*

rite $V_j = E(\pi_j)$. The pointwise selector inequality is $\mathcal{H}^{I(\pi_j)}[V_j] \geq \mathcal{H}^{\pi_j}[V_j]$. Comparison the assumed verification inequality imply $V_{j+1} \geq V_j$. Precompactness of the evaluation ge implies boundedness. Therefore at each state-time point the monotone sequence has a t $\bar{V}$.

very subsequence of V_j has a further convergent subsequence in the declared strong norm. niform function limit must be $\bar{V}$, by monotonicity and pointwise uniqueness of the limit. esequently the whole sequence converges uniformly to $\bar{V}$; in fact it converges in the declared ng norm because otherwise a subsequence separated from its unique possible strong cluster t would contradict precompactness. Derivatives of any strong cluster point are the derivatives

passing an evaluation equation to the limit. $\pi_{j_m} \rightarrow p$ along a policy subsequence. Continuity of evaluation gives $E(p) = \bar{V}$. Continuity at p and of E gives

$$E(I(p)) = \lim_m E(I(\pi_{j_m})) \stackrel{\text{NBO; R10-PRESERVATION SUPPLEMENT}}{=} \lim_m V_{j_m+1} = \bar{V}.$$

59

policy $I(p)$ attains the Hamiltonian maximum evaluated at $E(p) = \bar{V}$. Its evaluation at this value therefore gives

$$-\bar{V}_t = \mathcal{H}^{I(p)}[\bar{V}] = \sup_{a \in A} \mathcal{H}^a[\bar{V}].$$

boundary condition passes to the limit in the declared topology. Comparison identifies the value, and verification gives optimality. In particular $E(p) = V^*$, even if p and $I(p)$ differ. No step identifies successive limiting *policies* merely from a small difference of their

proposition concerns an exact operator on a compact, selector-closed policy class. Neither a neural network nor a finite-sample optimizer automatically meets those assumptions. The enacted finite policy is instead assessed by Theorem 2. For recursive drivers the comparison conditions have to be supplied for that driver and value domain; the proof does not obtain a conclusion from the word “recursive.”

A.2. Proof of Theorem 1

we discount Itô’s formula to the test function under any feasible policy a and stop at τ . Taking expectations yields the identity

$$v(t, s) - v(t, s) = \mathbb{E}_{t,s} \left[e^{-\rho(\tau-t)} (g - v)(\tau, S_\tau) + \int_t^\tau e^{-\rho(r-t)} R_a(v)(r, S_r) dr \right]. \quad (21)$$

Necessary, localization followed by the integrability hypothesis justifies removal of the boundary sequence. For the enacted policy, $R_\pi(v) \geq -\epsilon$ and the boundary difference is at least $J^\pi \geq v - b - C_\rho(T-t)\epsilon$. For every feasible alternative, $R_a(v) \leq R_\pi(v) + \eta \leq \epsilon + \eta$ and the boundary difference is at most b . Thus $J^a \leq v + b + C_\rho(T-t)(\epsilon + \eta)$. Taking a supremum over alternatives and subtracting proves (7). Attainment of that supremum is not

The proof does not need differentiability of the true value or differentiability of the policy with respect to its parameters. It does need a legitimate test function, a legitimate stopped process, and a boundary valid wherever any competing policy can travel. A residual measured under the current policy alone need not control an alternative policy. This is why an off-policy uniform bound, or an explicitly proved change-of-measure bound, is essential for the displayed worst-case bound.

0.1. Boundary and domain qualifications. The theorem bounds the stopped payoff as well as the continuation value, even when the optimal value lacks a classical extension to every boundary face. The diffusion can be degenerate in wealth under $\pi = 0$. We do not assert that all boundary faces are regular for every controlled diffusion, or that the continuous NDU value belongs to the boundary on all faces. The scalar payoff at a boundary starting state is settlement by definition; the continuation value may require a generalized viscosity boundary interpretation. The proof’s test-function and residual hypotheses would still have to be verified before numerical implementation. A finite boundary-switched critic is not used to bypass this requirement.

compact set and audit nodes form a d_h -net. Then

$$\|R\|_\infty \leq \max_j |R(s_j)| + L_R d_h.$$

60

same argument applies to a feasible improvement gap if its Lipschitz constant is established. The action coverage gives an additional $L_A h_A$ term only when a uniform action Lipschitz constant L_A is known. These inequalities are deterministic covering statements, not consequences of large sample size. No such global constants have been computed for the NDU neural critic, and no continuum certificate is reported for it.

A.3. Proof of Theorem 2

Positivity and row sums at most one imply that T_n^π and T_n are monotone and δ -Lipschitz in sup norm. Let $d_n^\pi = \|\hat{v}_n - V_n^\pi\|_\infty$. The evaluation defect gives $d_n^\pi \leq e_n + \delta d_{n+1}^\pi$, and hence

$$d_n^\pi \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} e_j.$$

For the optimal value, the residual with respect to T_n is at most $e_n + \eta_n$ in absolute value, because $\hat{v}_n - T_n \hat{v}_{n+1} = (\hat{v}_n - T_n^\pi \hat{v}_{n+1}) - (T_n \hat{v}_{n+1} - T_n^\pi \hat{v}_{n+1})$. The same backward estimate

$$\|\hat{v}_n - V_n^*\|_\infty \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

The triangle inequality proves the result. For exact re-evaluation, both e and b vanish; the code computes each date's maximum feasible gain and accumulates it with the factor δ .

Early liquidation creates no additional continuation coefficient: its payoff, including the one-step discount, is already in r_n^a . The substochastic surviving row sums remain at most one. Boundary nodes are replaced by their immediate settlement operator in both the optimizer and policy evaluator. The theorem therefore applies to the finite model even though its boundary policy outputs are not economically enacted.

A.3.0.1. Adding a verified numerical-economy error. Suppose, for the same policies and initial states, an independent argument establishes $|J^\pi - J_h^\pi| \leq \zeta$ uniformly and $|V^* - V_h^*| \leq \zeta_*$. Then $V^* - J^\pi \leq \zeta_* + (V_h^* - J_h^\pi) + \zeta$. This decomposition identifies the missing bridge without guessing it a zero. In particular, a coarse-fine value difference is neither ζ nor ζ_* without a finer error argument.

A.4. Proof of Theorem 3

For each admissible policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing in k , since $B \geq 0$. The supremum of these affine functions is convex; the pointwise supremum of nonincreasing functions is nonincreasing. At an initial state where optimizers exist, their optimality

$$\begin{aligned} A(\pi_1) - k_1 B(\pi_1) &\geq A(\pi_2) - k_1 B(\pi_2), \\ A(\pi_2) - k_2 B(\pi_2) &\geq A(\pi_1) - k_2 B(\pi_1). \end{aligned}$$

g multiple optimal policies. Moreover $-B(\pi_k)$ is a subgradient of V at k because $V(k) - (\tilde{k} - k)B(\pi_k)$. At a differentiability point this subgradient equals $V'(k)$. Finally restricted feasible policy set contains the set with $\theta = 0$, so its supremum is no smaller. Argument survives the killed finite approximation, the preservation of discounting, and θ do not depend on k , while the per-action cost remains exactly k times its discounted present exposure. It would not apply without modification if changing k also changed the action fee, the feasible action set, the utility normalization, or the law of preference shocks. Nothing about a pointwise ordering of θ , a portfolio-share ordering, or identification of θ from observed data.

NDIX B: GLOBAL ACTION SELECTION AND THE NDU STOCHASTIC OPERATOR

B.1. Constrained Hamiltonian maximization

fixed critic jet, the NDU maximand separates across c, θ, π . Let their bounds be (c_-, c_+) , (θ_-, θ_+) , and (π_-, π_+) . Consumption maximizes $c^{1-u}/(1-u) - V_X c$, whose second derivative $c^{u-1} < 0$. If $V_X > 0$, its optimum is the projection of $V_X^{-1/u}$ onto $[c_-, c_+]$. If $V_X \leq 0$, the objective is increasing and the optimum is c_+ . Adjustment maximizes $V_u \theta - k\theta^2/2$ and selects the projection of V_u/k . The portfolio objective as $a_1\pi + a_2\pi^2/2$, where

$$a_1 = .06XV_X + .2(.05)(-.25)XV_{uX}, \quad a_2 = .2^2X^2V_{XX}.$$

If $a_2 = 0$, select the projected value $-a_1/a_2$. If $a_2 > 0$, compare the two endpoints. If $a_2 = 0$, the endpoint indicated by the sign of a_1 , with any feasible action optimal when $a_1 = 0$. A nonconcave portfolio slice need not create an inaccessible local search problem in this model. The finite algorithm instead enumerates its declared action grid; it never labels any neural parameter as a global action optimum.

For a general differentiable concave maximand, concavity gives $H(a) \leq H(a_0) + \langle \nabla H(a_0), a - a_0 \rangle$. Taking a supremum proves (9). If an interior maximizer and an m -concave objective are known, the familiar gradient-square bound $H^* - H(a_0) \leq \|\nabla H(a_0)\|^2/(2m)$ follows by completing the square. The supporting-hyperplane formulation is useful at constrained optima. Neither argument transfers concavity from actions to shared parameters.

B.2. Local consistency and preference reachability

Let $\Delta S = (\theta, (r + e\pi)X - c)$ and let C denote the covariance matrix of the two-dimensional state. Before the exit correction, (16) satisfies

$$\mathbb{E}\Delta S = b\Delta, \quad \text{Cov}(\Delta S) = C\Delta, \quad C_{uX} = \sigma_u\sigma_X\pi X\rho_{uX}.$$

Polynomial tests $1, u, X, u^2, X^2, uX$ give the correct constant and linear generators and quadratic-generator errors $O(\Delta)$ from the drift outer product. At $(u, X) = (2, 1)$, $(\rho, \pi) = (.4, .13, .65)$, the continuous generators are

$$(0, .13, -.341, .5225, -.6651, -.553625).$$

Computable checks step sizes $.1, .05, .025, .0125$. The covariance error is below 10^{-12} and quadratic-generator discrepancy declines at the predicted rate. These tests include, rather than just the drift, the cross term. At a zero-adjustment interior state, the generator contribution of u^2 is not zero.

placement is $\theta\Delta$ and bilinear continuation varies between nodes. A smooth test function ϕ in u therefore sees every nonzero feasible adjustment even when $|\theta|\Delta$ is less than half a spacing. Preference reachability is not inferred from the existence of a control parameter in the data. 62

Positive bilinear interpolation on a smooth function has error of order $h_u^2 + h_X^2$ under appropriate derivative bounds. Dividing by Δ produces a generator contribution of order $(h_u^2 + h_X^2)/\Delta$. As an interior consistency limit requires a coupled choice such as $(h_u^2 + h_X^2)/\Delta \rightarrow 0$, together with vanishing action and time errors. Near the killed boundary, additional exit-time consistency and boundary regularity are needed. The four-branch linear-segment rule is not an exact Whittaker-bridge crossing calculation. These qualifications explain why the displayed separate elements are informative diagnostics but not a continuum error certificate.

B.3. Boundary cash flows and recorded exits

For a proposed increment d , the exit fraction is the minimum positive fraction at which $s + \lambda d$ crosses a domain face, capped at one. At a crossing the continuation value is $g(t + \lambda\Delta, s + \lambda d)$, the flow receives discount integral $[1 - \exp(-\rho\lambda\Delta)]/\rho$, and there is no continuation policy or settlement. A regression starts at $X = .51$, chooses $c = .8, \pi = 0$, and verifies that every path reaches $X = .5$ with fraction strictly below one and then terminates. The program never augments wealth to $.5$ and permits another consumption date under the pre-exit contract. The exit ledger distinguishes an exit event from a post-correction state. The adjustment budget is accumulated only until settlement. For a surviving branch interpolated to a boundary node, the t -date boundary settlement is part of the finite kernel. This interpolation effect is a numerical artifact, not a physical reflection. The settlement fee is a model primitive; comparing fees 6, 8, 10 is explicitly labeled a comparison of contracts.

APPENDIX C: RECURSIVE AGGREGATION, TEMPORAL SELVES, AND STRATEGIC INTERACTION

C.1. The recursive aggregator and its domain

For $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$, expand the aggregator as in (17). Its consumption derivative is

$$f_c(c, V) = \rho c^{a-1} (bV)^{1-a/b} > 0.$$

value derivative is

$$f_V(c, V) = \frac{\rho}{a} (b - a) c^a (bV)^{-a/b} - \frac{\rho b}{a}.$$

On compact ranges with consumption bounded away from zero and bV bounded away from zero, these derivatives are bounded. This is a local Lipschitz conclusion on the specified range, not a global sign or existence theorem for all values. The transform $V = e^W/b$ enforces $bV > 0$, but cannot by itself verify integrability, terminal compatibility, comparison, or transversality. The policy-wise recursive evaluation solves (3); in the Markov setting its equation is $v_t + \mathcal{L}^\pi v + \epsilon_t = 0$. When f is Lipschitz in v with constant L , standard comparison of a perturbed equation with a verified evaluation gives a finite-horizon error envelope of the form

$$e^{L(T-t)} b + \int_t^T e^{L(r-t)} \epsilon dr$$

C.2. Executed implicit recursive reference

Supplementary calculation fixes $\gamma = 5$, $\psi = 1.5$, and $\rho = .04$ on a deterministic saving with ten steps of length .05. At each wealth node $x \in [.5, 2]$, consumption is $c = mx$ with m in $[\frac{1}{10}, \frac{1}{6}]$, and the next wealth is the stated saving transition in the code. For the scalar evaluation equation is

$$y = I_h V_{n+1}(x') + \Delta f(c, y), \quad \text{by } y > 0.$$

The negative-value root is bracketed and solved numerically before maximizing over the action. The terminal target is x^b/b . Outside the wealth lattice, continuation uses the specified homothetic tail as numerical boundary data at that date. It is not a solution on an unbounded wealth domain. At the central node the recorded value is $-.29658656$ and the consumption fraction is the lower bound .01. This active bound is reported rather than used as an interior stationary ratio.

Sign-domain and implicit-equation residual tests are actual evaluated quantities. Neural error is null because this experiment solves scalar implicit reference equations rather than training a recursive neural policy. The general recursive portfolio model remains part of the operator formulation and analytical derivations; its stationary ratios .3 and .0455 remain as anchors until their full infinite-horizon verification is supplied.

C.3. Ordinary continuation and acting-self utility

With a fixed continuation policy, ordinary log-wealth utility in the two-consumption-date case has form $V_n(w) = A_n \log w + B_n$. Starting from $A_N = 1, B_N = 0$, the sophisticated recursion uses

$$\frac{1}{1 - \beta \delta A_{n+1}}, \quad c_n = \alpha_n w, \quad A_n = 1 + \delta A_{n+1}, \quad B_n = \log \alpha_n + \delta [A_{n+1} \log(1 - \alpha_n) + B_{n+1}].$$

Improvement uses $\beta \delta$. With $\delta = 1$ and two decisions, $A_1 = 2$ and $B_1 = \log(1 + 2\beta)$. Substituting β into the evaluation coefficient instead produces the different form $1/(1 + \beta + \beta^2)$. At $\beta = 1$ the two formulas coincide, so this special case cannot distinguish the distinction.

A one-shot deviation is evaluated against the same frozen continuation policy whose equilibrium is being claimed. Replacing its continuation with a different grid's optimized value changes the result. For a general finite sophisticated-self computation, all feasible actions can be evaluated at each date using the correctly evaluated future policy. For continuous actions a search gives only a feasible deviation lower bound unless concavity, a global oracle, or an upper-bound calculation closes the search gap.

C.4. Dynamic games and equilibrium selection

$J_i(\pi_i, \pi_{-i})$ denote player i 's payoff with its own objective and transition law. A Nash equilibrium satisfies the bounds

$$\sup_{\tilde{\pi}_i} J_i(\tilde{\pi}_i, \pi_{-i}) - J_i(\pi_i, \pi_{-i}),$$

lateral derivative graph. In the static benchmark $q_i(1 - q_i - q_j)$, the derivative with a detached is $1 - 2q_i - q_j$. It vanishes at $q_i = q_j = 1/3$, whereas joint optimization selects $1/4$ and es a profitable deviation. The executable graph regression verifies both the own gradient and absence of a rival parameter gradient.

ne three-date capacity game has an exogenous Markov demand state and endogenous future city. At each state-date pair, enumerate the two players' current feasible output–investment ons using continuation values from the already selected future equilibria. A pure equilibrium action pair that is a best response for each player against the other current action and the d future policy profile. Every enumerated state has at least one pure equilibrium in this ticular game. Lexicographic tie selection is recorded, and the full profile includes asymmetric ces. In a finite-horizon game, this backward best-response construction establishes subgame ection on the finite state space; no uniqueness follows from it.

or large-population or competitive-limit analysis, the demand and capacity scaling, invest- t technology, and equilibrium concept must be fixed before taking a limit. Symmetric alization is neither a proof of equilibrium nor an equilibrium-selection theorem. Those tions are retained as model extensions rather than filled with the historical illustrative eries. No claim about a trained high-dimensional game or a solved competitive limit is hed to the current exact finite game.

APPENDIX D: TRACE ESTIMANDS AND COMPUTATIONAL COST

or standard Gaussian z and symmetric A , diagonalization and the independent squared- nal moments give $\mathbb{E}z'Az = \text{tr } A$ and $\text{Var}(z'Az) = 2\|A\|_F^2$. Independence across K probes ds variance $2\|A\|_F^2/K$ for their mean. Expanding the squared residual then proves (20). Rademacher probes the diagonal contribution to variance vanishes, so the formula must be aged rather than reused uncritically.

or $K \geq 2$, define the cross-product correction

$$U_K = a^2 - a\bar{q}_K + \frac{1}{4K(K-1)} \sum_{i \neq j} q_i q_j. \quad (22)$$

pendence of distinct probes gives $\mathbb{E}U_K = (a - \text{tr } A/2)^2$. The statistic is unbiased but can egative in a finite batch and may have a different variance and optimization behavior. therefore an additional estimand, not a blanket replacement justified by unbiasedness e. The audit reports its empirical mean and standard error separately. When A depends on ork parameters, the uncorrected variance term itself varies with those parameters; a squared hastic trace can consequently change the expected minimizer.

et C_{jet} denote the actual cost of a directional second derivative and its parameter gradient for chosen network. If Σ has m columns, exact contraction can be organized as m directional nd derivatives, with loading and batching costs. A K -probe contraction costs roughly KC_{jet} those costs; it is not an unconditional $O(d)$ solver. Total work additionally multiplies by ocation samples, training iterations, and any restarts. Peak memory, warm-up, and accelerator hronization must be measured for a hardware comparison. The current CPU timers do not sure peak memory, so that metric remains null.

0.0.1. *The coupled dynamic regression.* For the finite-horizon quadratic-cost economy, policy $a = Wx$ gives a quadratic value $x'Px + c$. Given a frozen next evaluator, the actor mizes the expected quadratic stage-plus-continuation cost. Gaussian shocks contribute

ed against the frozen evaluator by L-BFGS, while an independent Riccati recursion
 es the benchmark. Both use identical A, Q, R, S , horizon and terminal cost. Correlation
 e-transition covariance here, not a metadata label on a static objective.
 nown quadratic family makes this a favorable regression. **APPENDIX E**, and
 Ricci times are approximately .00060, .00040, .00043 seconds and actor times are
 323, .0384 seconds. The first timing includes startup effects. These numbers neither
 a dimension scaling law nor compare against adaptive sparse grids. The original static
 quadratic program is retained in the R3 archive as a small linear-algebra check, separate
 s dynamic experiment.

APPENDIX E: APPROXIMATION, COUNTEREXAMPLES, AND OPTIMIZATION LIMITS

E.1. A composite objective can bias evaluation

der a one-state terminal evaluation problem whose true value is zero and whose critic is
 v . The exact evaluation residual is v . A joint surrogate $v^2 - \lambda v$ has minimizer $v = \lambda/2$,
 evaluated value, for every $\lambda \neq 0$. The example isolates the logical issue: optimizing
 nce and evaluation over the same value parameter need not preserve the evaluation
 . Adding a positive residual weight changes the bias but does not prove exact evaluation.
 arated actor has no gradient into that value parameter, so it does not create this particular

arly, zero-centering a residual is not a stability theorem. The derivative of $R^2/2$ is
 nd its Jacobian, conditioning, sampling, and step sizes govern a finite optimizer. Value
 nce in a recursive driver remains inside this derivative. A zero target does not make the
 e independent of the value level.

E.2. A small squared residual does not select a viscosity solution

-1, 1) with boundary values zero, consider $|v'| = 1$. The nonnegative distance function
 s the viscosity solution. The function $|x| - 1$ has the same boundary values and satisfies
 almost everywhere, but it fails the viscosity supersolution test at zero: the constant test
 -1 touches it from below and has $|\phi'| - 1 = -1 < 0$.
 th approximations $v_\varepsilon(x) = \sqrt{x^2 + \varepsilon^2} - \sqrt{1 + \varepsilon^2}$ have exactly the boundary data and
 e uniformly to $|x| - 1$. Their squared residual $\int_{-1}^1 (|v'_\varepsilon| - 1)^2 dx$ tends to zero by domi-
 nvergence, although the limit is not the viscosity solution. Thus boundary agreement,
 networks, and an L^2 residual do not establish viscosity selection. Monotone, stable,
 nt schemes with appropriate comparison can provide a different route (Barles and
 dis, 1991); positivity of an interior interpolation alone does not verify all of those
 ses at a killed boundary.

th nonpolynomial neural networks can approximate a sufficiently smooth function and
 ves on compact subsets under the relevant approximation hypotheses. This does not
 derivatives at a kink, ensure global boundary regularity, close a finite policy class under a
 izing selector, or prove that stochastic optimization finds the approximant. The retained
 ve-approximation motivation is used only in that conditional sense.

E.3. Two-timescale statements and suboptimal attractors

rojected stochastic approximation, a conventional ODE analysis requires bounded
 suitable continuity and stability, martingale-difference noise with bounded conditional

step-size ratio, plus a globally attracting critic equilibrium for each frozen actor. Under the appropriate additional technical conditions, the actor limit set is governed by the corresponding limiting ODE and its internally chain-transitive sets. This is not a global optimality statement. For $H(a) = -(a^2 - 1)^2 + .2a$, the negative local maximizer is an asymptotically stable isolated stationary point with strictly positive global improvement gap. The audit solves the cubic stationarity equation and evaluates the gap .3998745872. Exact critic evaluation and zero noise do not remove that attractor. To conclude optimality one must establish, for the actual actor dynamics, that every relevant limiting set has zero feasible improvement gap, or use an applicable global concavity or gradient-domination argument. The runs use Adam and L-BFGS and are assessed by their measured errors; they are not examples of a diminishing-step asymptotic theorem.

APPENDIX F: IMPLEMENTATION, EVIDENCE CLASSES, AND REPRODUCIBILITY

F.1. *Executed parameterizations and data flow*

The continuous Merton critic has one learned log-amplitude and its actor has two learned policies. It uses float64 automatic differentiation through the value with respect to wealth, including the Hessian; critic and actor optimizers are separate. Its independently evaluated policy has a positive transversality decay D . Analytical controls are used only after training to measure error. The NDU run is different: time-specific nonlinear networks represent a finite Bellman operator. The actor maps two normalized states to 125 logits through two 48-unit tanh layers. Its critic has six features, including log distances to boundary faces, through two 48-unit tanh layers to allow adjustment to terminal utility. Next-date critics are frozen, the actor target uses only their continuation predictions, and the current critic is fitted to the enacted actor’s policy-evaluation output. The independent NumPy optimizer and evaluator are not imported as training labels. Training residuals are not out-of-sample certificates because all interior finite states are location points. The independent evaluator changes the computational path, not the economic model. The source contains separate NumPy and PyTorch implementations of the transition and interpolation, whose equality is tested on deterministic data. Cross-grid reference calculations and discretization diagnostics; they have not been relabeled a held-out neural accuracy result.

F.2. *Interpreting metrics*

A value error against the finite optimal reference, a critic error against the evaluated policy, a case-fine value difference, and a unilateral gain are different estimands. Policy component differences are retained for consumption, adjustment, and portfolio share. Boundary-node actor outputs are not enacted after settlement; full-grid maxima and interior summaries must be distinguished when interpreting them. A finite exhaustive best-response maximum is a finite-level certificate. A feasible search over only some continuous deviations is merely a lower bound on exploitability. Execution status and tolerance status are separate fields. A program that completes can miss a target; a timed-out audit has no final solution certificate. A null numerical error has an explicit origin, such as an unmeasured continuous-diffusion error or a recursive reference that did not converge to a neural policy. A numerical zero is used only for a quantity actually calculated, such as a finite game deviation maximum or a tested monotonicity violation. The initial failed neural source and its weights, metrics, and raw arrays are retained in the full review bundle. The successful three-seed sources and weights are separate. The attempted next-date runs are marked incomplete. Wall-clock times are metadata and are excluded from

identifies the historical base, not a fictitious commit containing the subsequently created
The final manifest maps every delivered source to its actual file hash and repository path.

F.3. *Retained material and revision lineage*

NBO: R10.PRESERVATION SUPPLEMENT

67

new authoritative entry is `ECTA_R4.tex`. The original `ECTA.tex`, the R3-authoritative
`R2.tex`, its standalone supplement, embedded arrays, earlier reports, and R3 numerical
are left unchanged. Core model formulations, constrained policy derivations, recursive
tion, temporal-self and game extensions, trace analysis, and theoretical qualifications
on the current text and these appendices. Historical unexecuted plots are retained as
objects, not presented again as learned current outcomes. The retention map identifies
source locations and the corresponding corrected discussion. This preserves the scientific
without repeating incompatible equations as simultaneous current specifications.

The tables below reproduce all six delivered R5 TeX tables without changing their entries. We refer to the source and protocols reviewed on September 21, 2026. The R8 main paper gives the current interpretation: finite candidate certificates are not continuous certificates; safeguarded rules are distinct from raw rules; the in-house PINN-PI comparison is not the new author-code comparison; and these historical extension panels have only two seeds. The execution data, raw counts, checkpoints, and failed or incomplete attempts remain in the corresponding revision histories.

TABLE I
RETAINED R5 NDU PANEL

Configuration	n	Raw	Safeguarded	Max. bound	Seconds
baseline	10	10	10	0.0496	18.19
longer horizon	3	0	3	0.0064	34.64
width 32	2	1	2	0.0471	16.86
width 64	2	2	2	0.0461	20.03
lr 0.002	2	0	2	0.0050	18.52
lr 0.008	2	2	2	0.0339	17.95
actor steps 300	2	0	2	0.0065	13.66
critic steps 400	2	1	2	0.0440	15.70

In this table, n denotes the number of runs, not the number of time steps. Pass counts use the historical finite-candidate target. Seconds are the historical recorded training measure; this is not the new full safeguard re-audit timing.

TABLE II
RETAINED R5 QUADRATIC-CONTROL COMPARISON

d	Method	Mean loss	Sampling upper	Train (s)	Audit (s)	MiB
8	NBO	0.0010	0.0018	14.93	3.45	377.5
8	PINN-PI	-0.0001	0.0007	12.59	6.61	376.9
16	NBO	0.0037	0.0055	14.29	3.84	378.6
16	PINN-PI	0.0013	0.0032	11.67	5.69	377.7
32	NBO	0.0102	0.0145	16.29	3.72	381.3
32	PINN-PI	0.0075	0.0117	13.34	5.64	382.1

This is the two-seed Cole-Hopf/QMC benchmark with an analytic greedy control and an in-house PINN-PI comparator. The small negative estimated loss reflects reference or simulation uncertainty, not performance exceeding the true optimum. NBO has higher mean loss in each row pair. This result is retained despite the favorable R8 nonconvex-control comparison.

N	Endpoint value	Bridge value	Action sup error	Policy-loss bound
16	0.0392	0.0581	0.3783	0.1691
32	0.0616	0.0308	0.4653	0.0757
64	0.0623	0.0159	0.5033	0.0366
128	0.0538	0.0080	0.5346	0.0219
256	0.0436	0.0040	0.5502	0.0106

NBO: R10 PRESERVATION SUPPLEMENT

orical one-dimensional solver has an analytically manufactured verification pair and no trained actor. It is distinct from the new learned NDU-geometry test, which also explicitly modifies the running payoff.

TABLE IV
RETAINED R5 COUPLED NDU REFINEMENTS

N	State grid	n_a	h^2/Δ	V_0	B_0	Exit
12	13×16	3	0.2133	-1.624355	0.012266	0.1543
24	25×31	3	0.1067	-1.510132	0.012248	0.1558
48	49×61	5	0.0533	-1.391646	0.011969	0.1024
96	97×121	5	0.0267	-1.353683	0.012234	0.0987

iscretization factors change in this historical panel. The table supplies numerical differences, not validated continuous error bounds or a separated convergence order.

TABLE V
RETAINED R5 RECURSIVE-UTILITY POLICIES

Seed	Raw log loss	Final bound	Final log loss	m_0	p_0
51	0.0425	0.0000	0.0000	2.506	0.955
52	0.0502	0.0000	0.0000	2.442	1.093

raw neural policies miss the historical .01 log-value target. The final candidate-set bounds use repeated 961-action safeguards; the raw failures are not relabeled as neural successes.

TABLE VI
RETAINED R5 NEURAL GAME

Seed	Player 1, $t = 0$	Player 2, $t = 0$	All subgames	Seconds
61	0.00245	0.00673	0.00673	13.87
62	0.00312	0.00305	0.00368	14.04

a small finite-state, three-date game with exact continuous best-response auditing for the declared finite operator, not a continuous-state equilibrium scalability result.

W., G., and P. E. Souganidis (1991): “Convergence of Approximation Schemes for Fully Nonlinear Second Order Elliptic Equations,” *Asymptotic Analysis*, 4, 271–283. [40]

W., D. B., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58(4, Part 1), 785–801. DOI: 10.1287/opre.1090.0796. []

W., D. B., and J. E. Smith (2011): “Dynamic Portfolio Optimization with Transaction Costs: Heuristics and Dual Bounds,” *Management Science*, 57, 1752–1770. DOI: 10.1287/mnsc.1110.1377. []

W., J., and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85, 1575–1612. [24]

W., S. Fang, and T. Zhou (2025): “SOC-MartNet: A Martingale Neural Network for the Hamilton–Jacobi–Bellman Equation without Explicit $\inf_{u \in U} H$ in Stochastic Optimal Controls,” *SIAM Journal on Scientific Computing*, 47(4), C95–C819. DOI: 10.1137/24M1681033. [3, 14]

W., S. Fang, and T. Zhou (2025): “SOC-MartNet: A Martingale Neural Network for the Hamilton–Jacobi–Bellman Equation without Explicit $\inf H$ in Stochastic Optimal Controls,” arXiv:2405.03169, version 4, March 15. Historical preprint citation retained for the preservation supplement. [24]

W., D., and L. G. Epstein (1992): “Stochastic Differential Utility,” *Econometrica*, 60, 353–394. [15, 24]

W., L. G., and S. E. Zin (1989): “Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework,” *Econometrica*, 57, 937–969. [15, 24]

W., J., and W. E (2016): “Deep Learning Approximation for Stochastic Control Problems,” arXiv:1611.07422. [24]

W., J., A. Jentzen, and W. E (2018): “Solving High-Dimensional Partial Differential Equations Using Deep Learning,” *Proceedings of the National Academy of Sciences*, 115, 8505–8510. [24]

W., S. D., and A. Mijatović (2015): “On the Policy Improvement Algorithm in Continuous Time,” arXiv:1509.09041. [24]

W., K. L. (1998): *Numerical Methods in Economics*. MIT Press. [24]

W., Y., N. Cho, M. Kim, and Y. Kim (2026): “Physics-Informed Approach for Exploratory Hamilton–Jacobi–Bellman Equations via Policy Iterations,” *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(27), 22609–22616. DOI: 10.1609/aaai.v40i27.39421. [2, 14]

W., Y., M. Kim, Y. Kim, and N. Cho (2026): “Physics-Informed Policy Iteration for High-Dimensional Hamilton–Jacobi–Bellman Equations: Interior Error Bounds without Boundary Data,” arXiv:2508.01718, version 2, August 9. [14]

W., Y., Y. Kim, M. Kim, and N. Cho (2025): “Neural Policy Iteration for Stochastic Optimal Control: A Physics-Informed Approach,” arXiv:2508.01718, version 1. Historical citation retained for the preservation supplement; see updated 2026 entry for the current title and author order. [23]

W., J., H. J., and P. G. Dupuis (2001): *Numerical Methods for Stochastic Control Problems in Continuous Time*, second edition. Springer. DOI: 10.1007/978-1-4613-0007-6. []

W., J., J. B., D. Henrion, C. Prieur, and E. Trélat (2008): “Nonlinear Optimal Control via Occupation Measures and HJB-Relaxations,” *SIAM Journal on Control and Optimization*, 47, 1643–1666. DOI: 10.1137/070685051. []

W., J. Y., and Y. Kim (2025): “Hamilton-Jacobi Based Policy-Iteration via Deep Operator Learning,” *Neurocomputing*, 565, 130515. DOI: 10.1016/j.neucom.2025.130515. [3, 14]

W., J. Y., and Y. Kim (2024): “Hamilton-Jacobi Based Policy-Iteration via Deep Operator Learning,” arXiv:2406.10920. Historical preprint citation retained for the preservation supplement. [24]

W., J., R. C. (1969): “Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case,” *Review of Economics and Statistics*, 51, 247–257. [24]

W., J. (2025): “Neural Foundations of Endogenous Risk Aversion,” unpublished manuscript, Peking University; motivating manuscript identified in the retained NBO source. [24]